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Structure-based Druggability Predictions of RNA Binding Sites and Design of RNA Ligands

Doctoral thesis defence

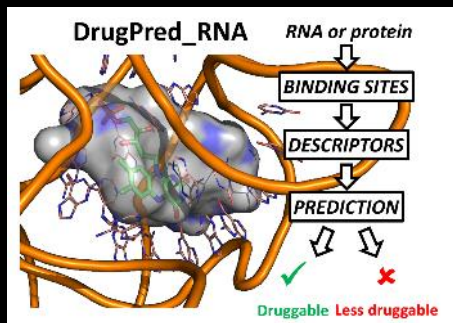
Illimar Hugo Rekand (He/him)

Brenk group, Department of Biomedicine

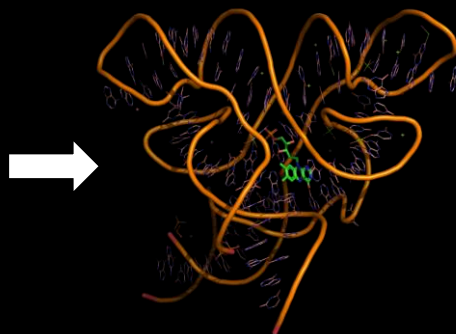
Haug group, Department of Chemistry

UNIVERSITY OF BERGEN

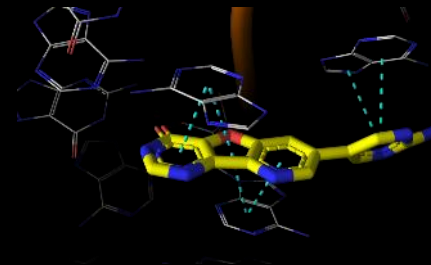




Assessment of druggability
properties of RNA binding
sites



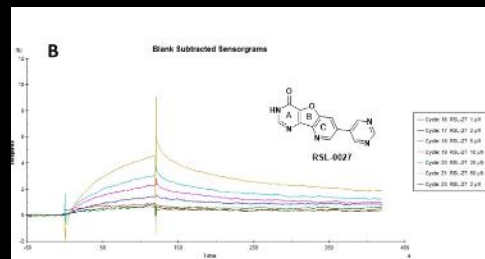
Riboswitches & their
relevancy as antibiotic
targets



Ligand design for the
FMN riboswitch



Synthesis of proposed
ligands



Assay results &
evaluation



**How do we know which
targets are good candidates
for drug development?**



Drug development is extremely resource demanding

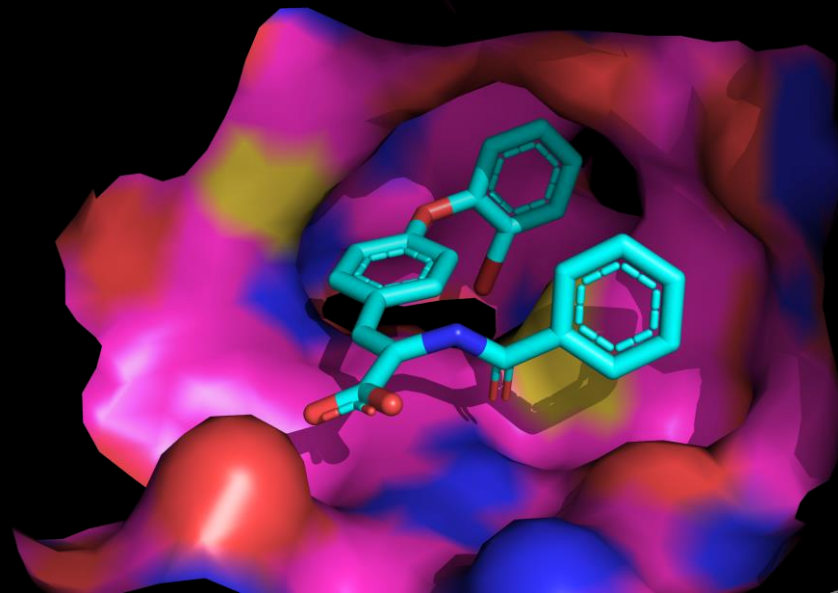
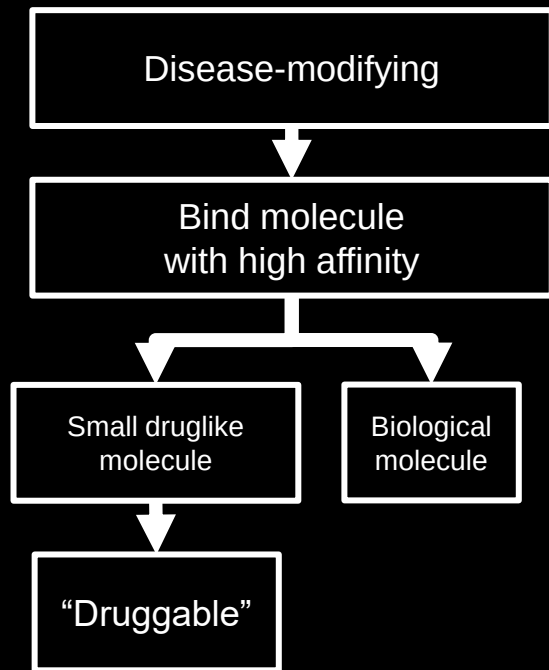
- Time: ~12 years
- Money: \$1.3 billion per approved drug
- >90% failure risk
- Which horse should one bet on?
- Where do these projects fail?
- Why develop a drug for a target which is ill suited?



By Hippodrome de Deauville – Clairefontaine ©, Attribution,
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What criteria should we ask from a drug target?

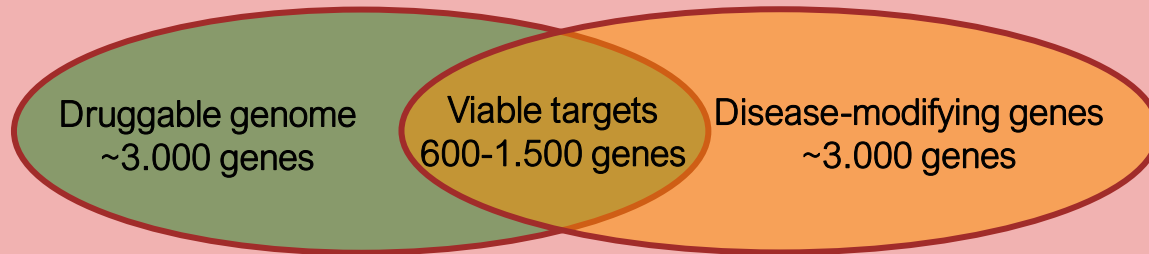


Hepatitis C virus RNA polymerase complex with inhibitor (PDB: 1yvf)



Druggability: Genomic approach

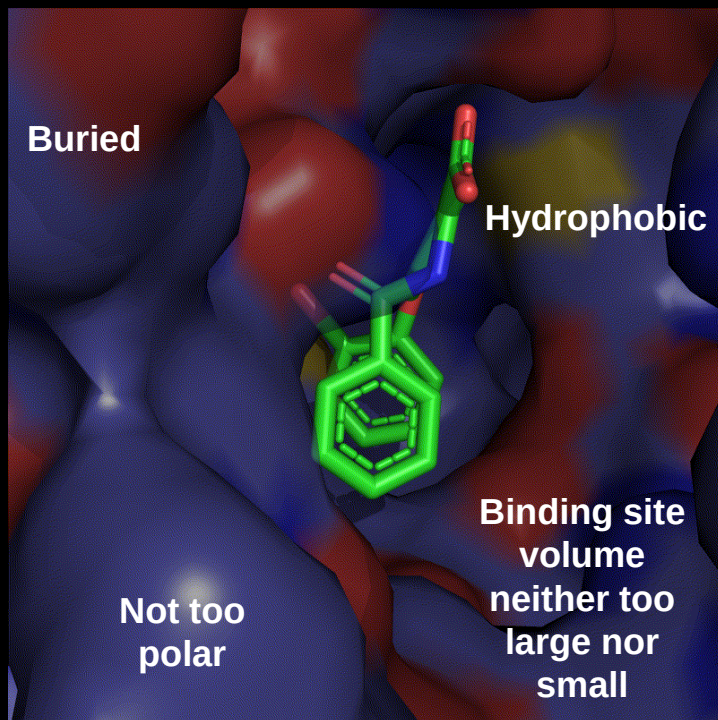
The human genome:
~30.000
genes



Limited number of potential protein
targets

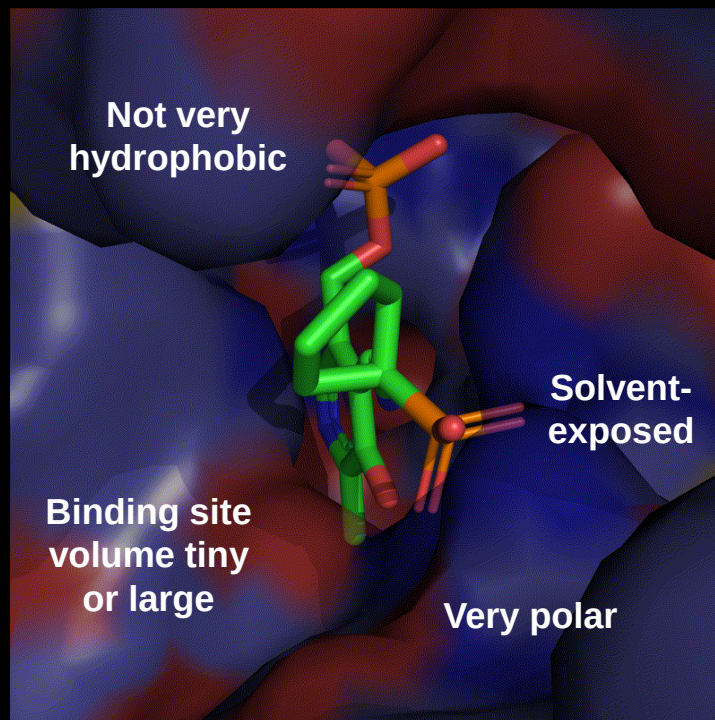


Druggable



Hepatitis C virus RNA polymerase complex with inhibitor (PDB: 1yvf)

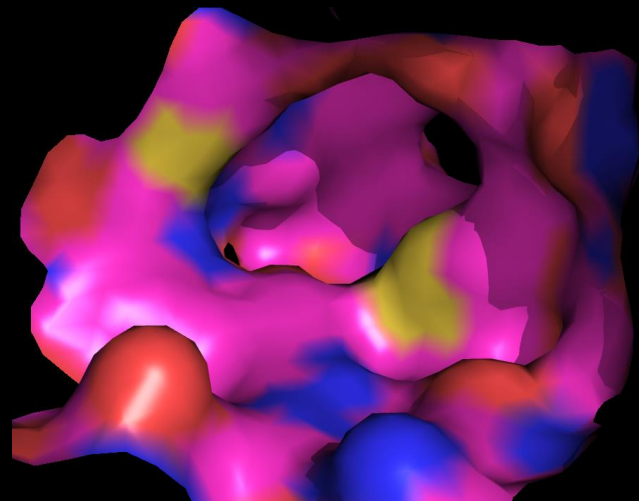
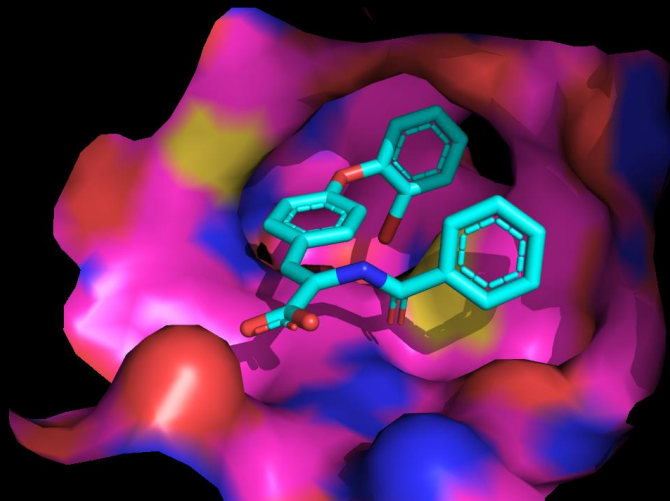
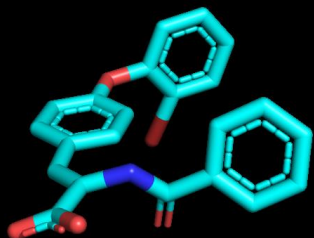
Less druggable



Dialkylglycine decarboxylase complex with inhibitor (PDB: 1m0n)



Druggability vs. druglikeness



Druglikeness of the ligand

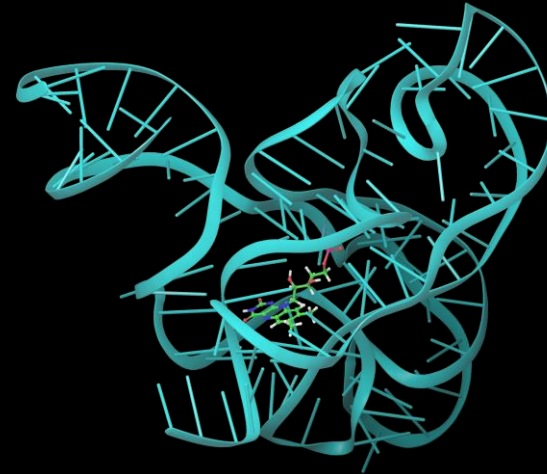
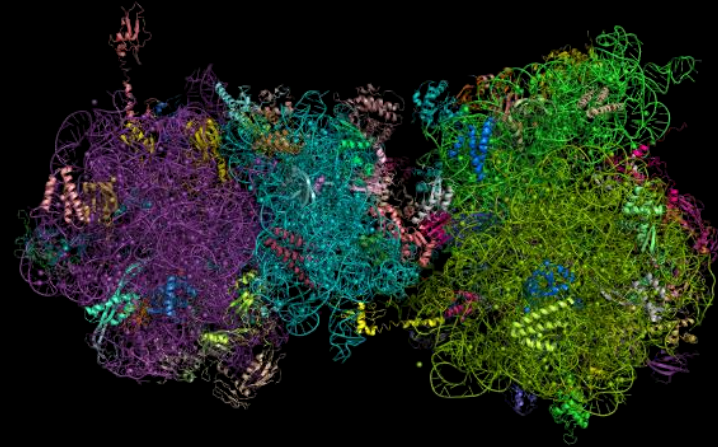
Ligand-binding site complex

Druggability properties
of the binding site



RNA as a drug target

- Folded RNA can create binding pockets
- Small molecules
- Examples:
Ribosomes
Riboswitches

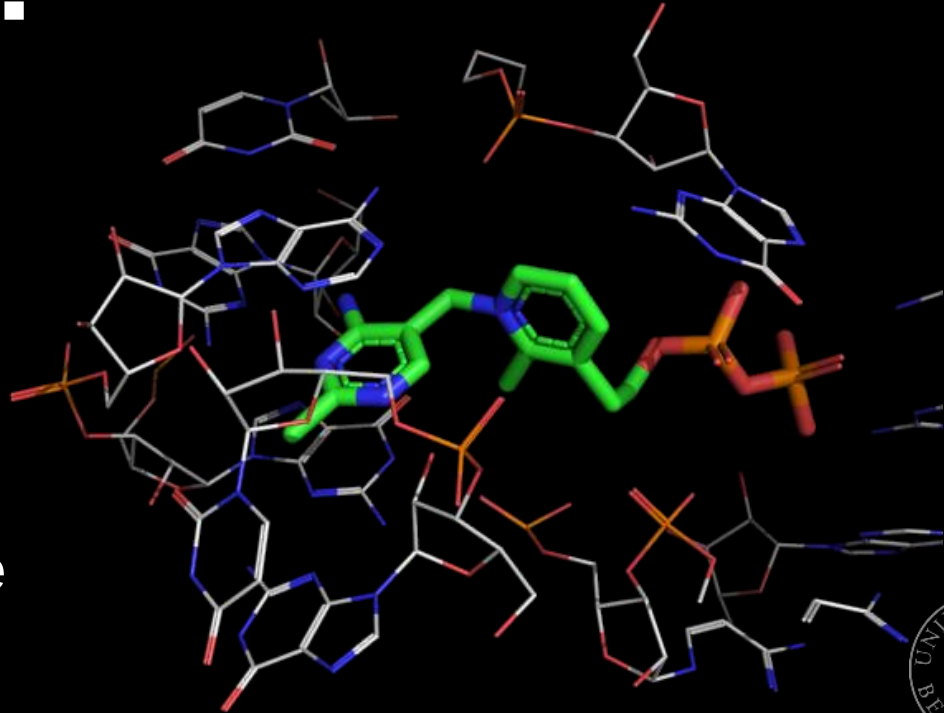


FMN riboswitch (2yie)



But isn't RNA...

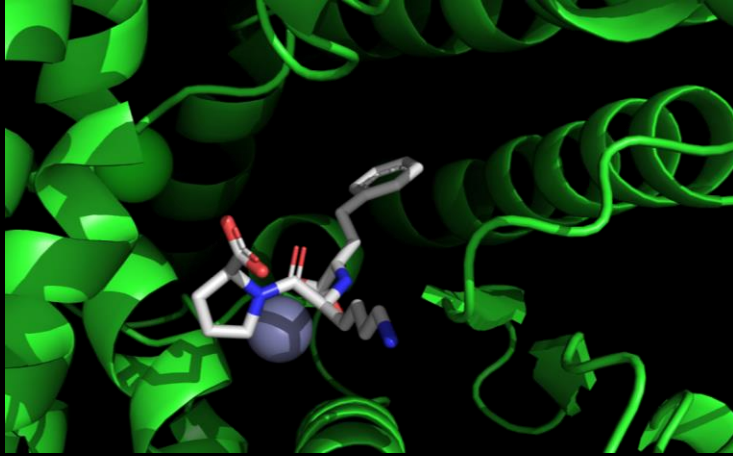
- “Too unstable?”
Protein database (PDB):
Containing RNA: 4700
Only RNA: 1400
- “Too polar?”
Nucleic bases provide
hydrophobic surface



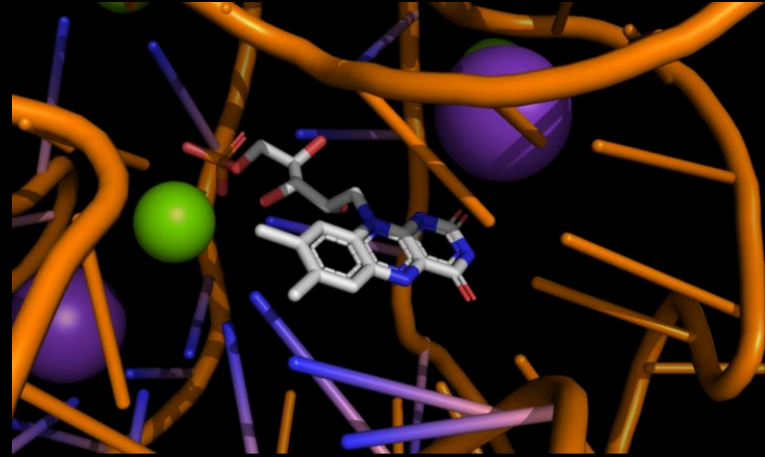
TPP riboswitch (PDB code: 3d2v)



Protein vs. RNA – what to consider?



Human Angiotensin Converting
Enzyme in complex with lisinopril.
(1o86)



FMN riboswitch in complex with
FMN (2yie)

- 20 amino acids vs. 4 residues
- Basic interactions principles apply

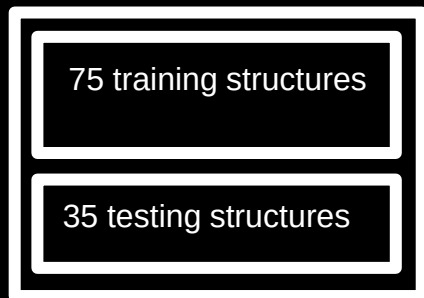


DrugPred_RNA: A druggability predictor compatible with RNA

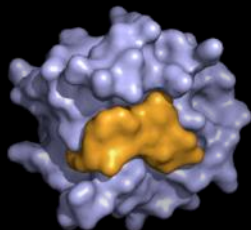


DrugPred: a druggability prediction software

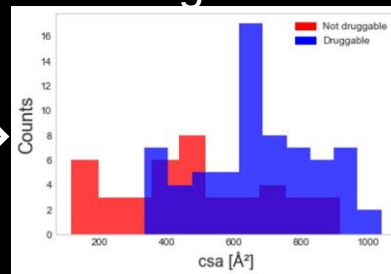
110 non-redundant known druggable or less druggable protein structures



Data preparation

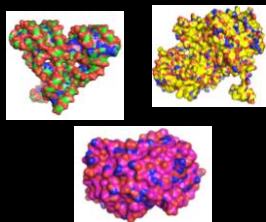


Binding site data



Evaluate and improve

Machine learning model



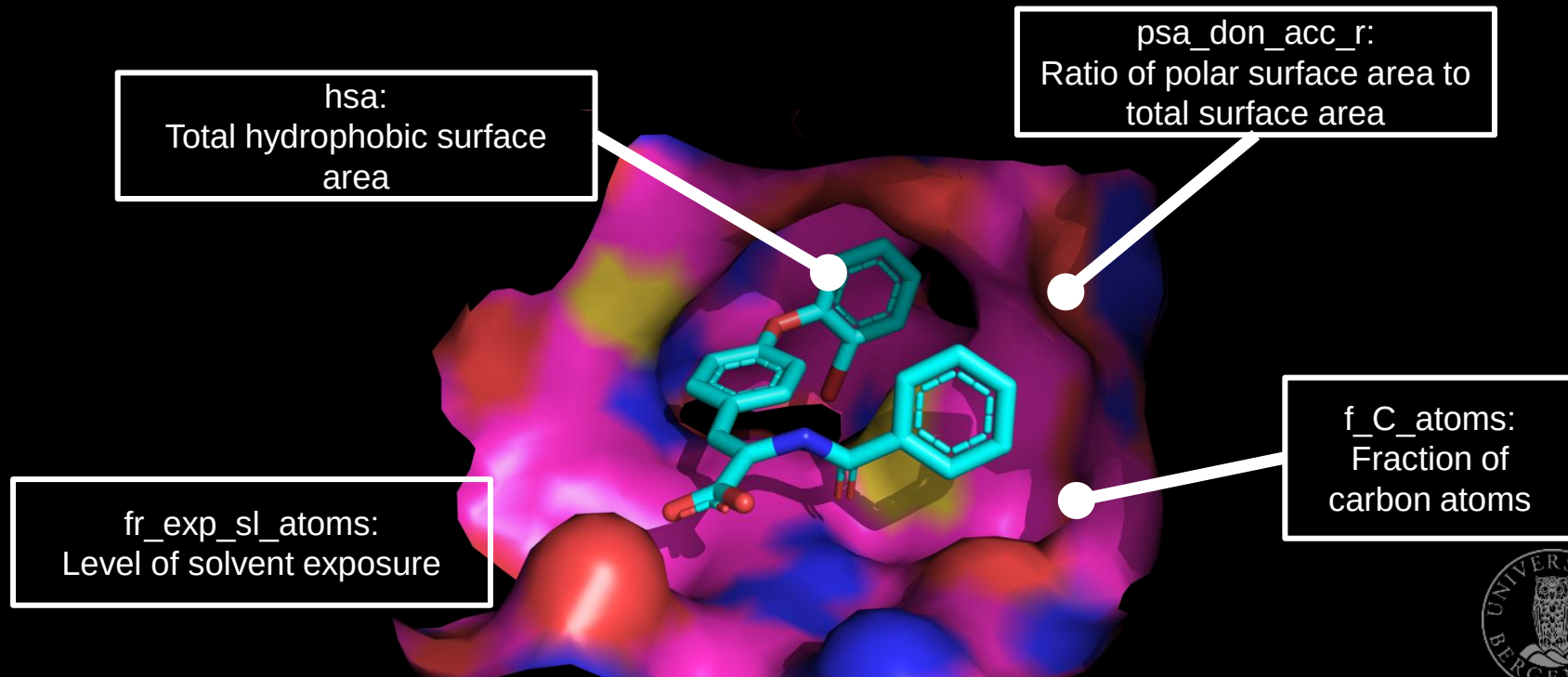
Data preparation

Calculate binding site properties

Machine learning model

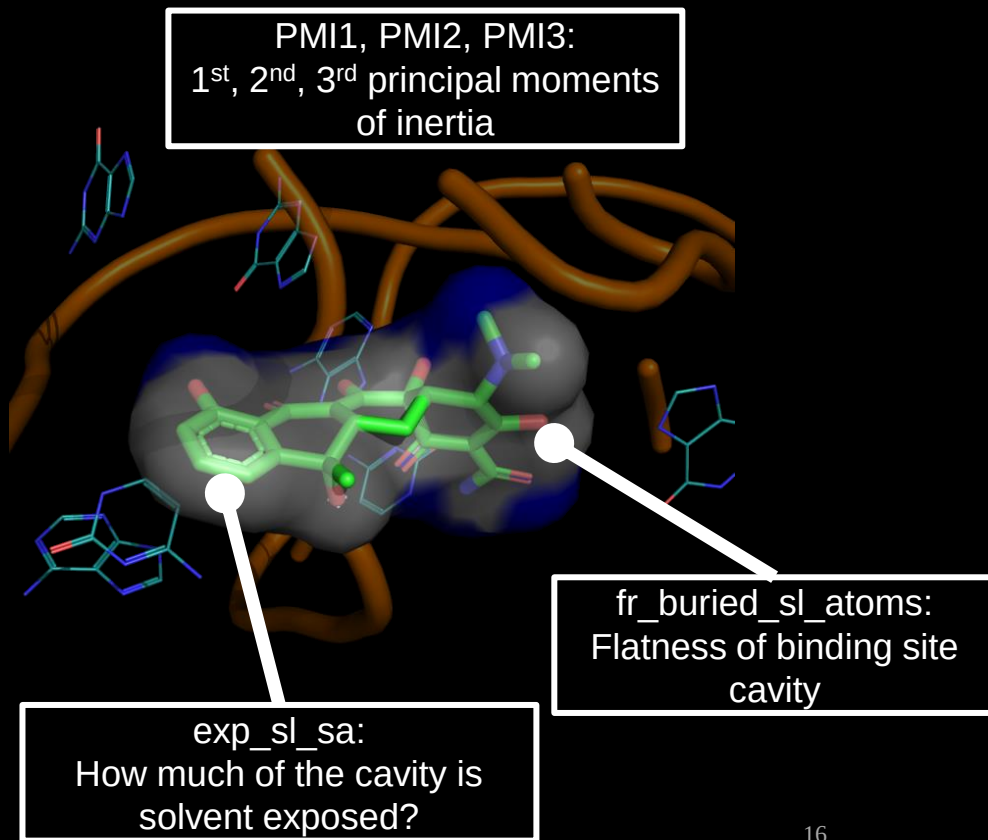


Some binding site descriptor examples

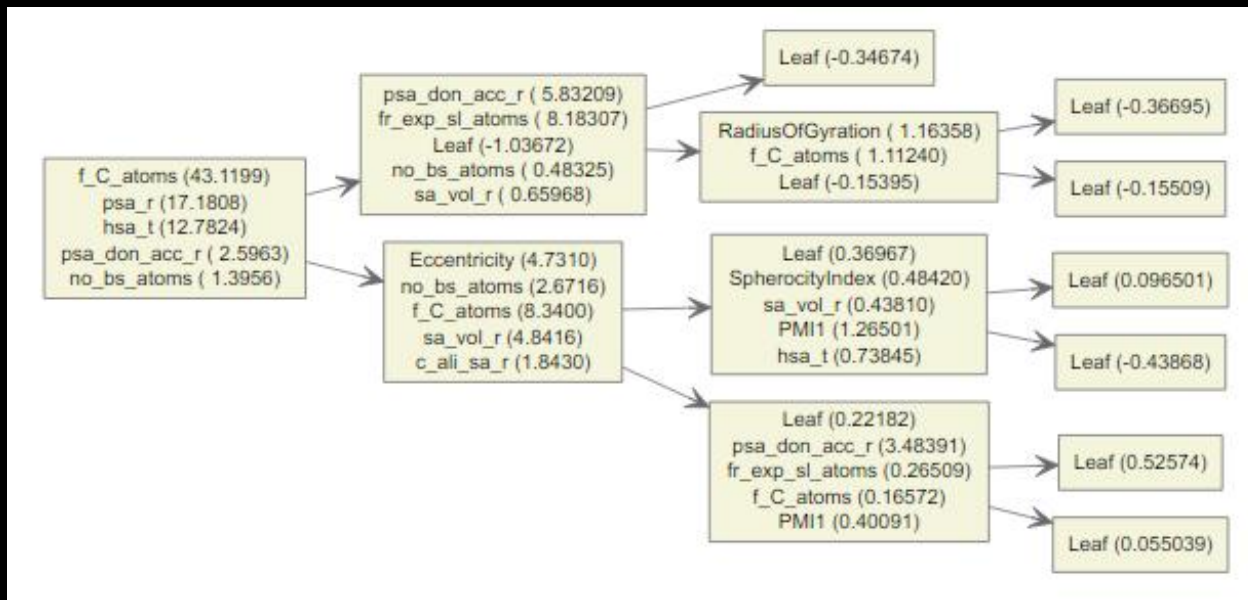


DrugPred was developed for protein binding sites

- Certain high-ranking descriptors were only compliant with protein structures
- Some descriptors removed, some new added
- Still trained on protein dataset
- New prediction method

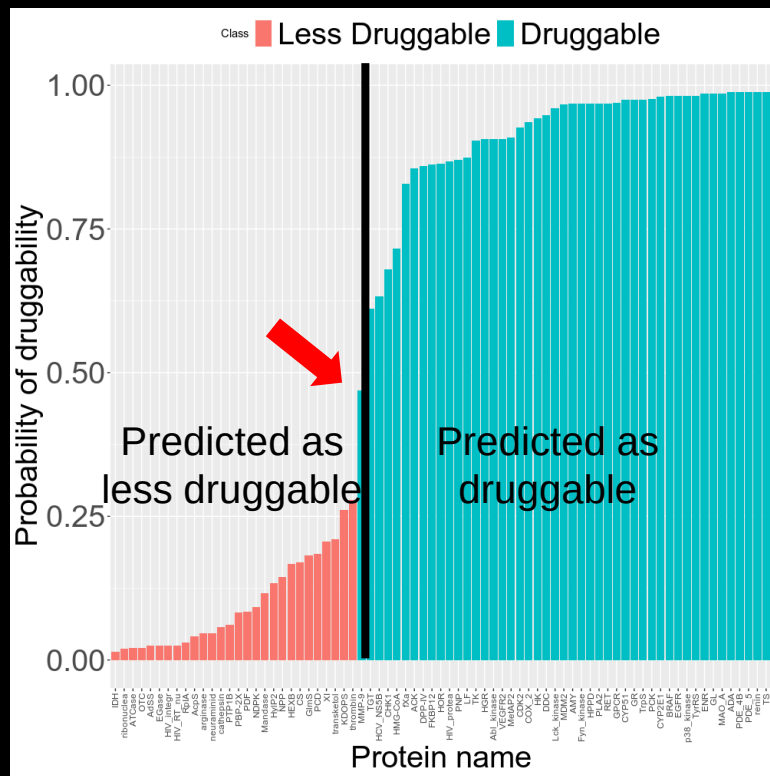


Decision tree-based machine learning

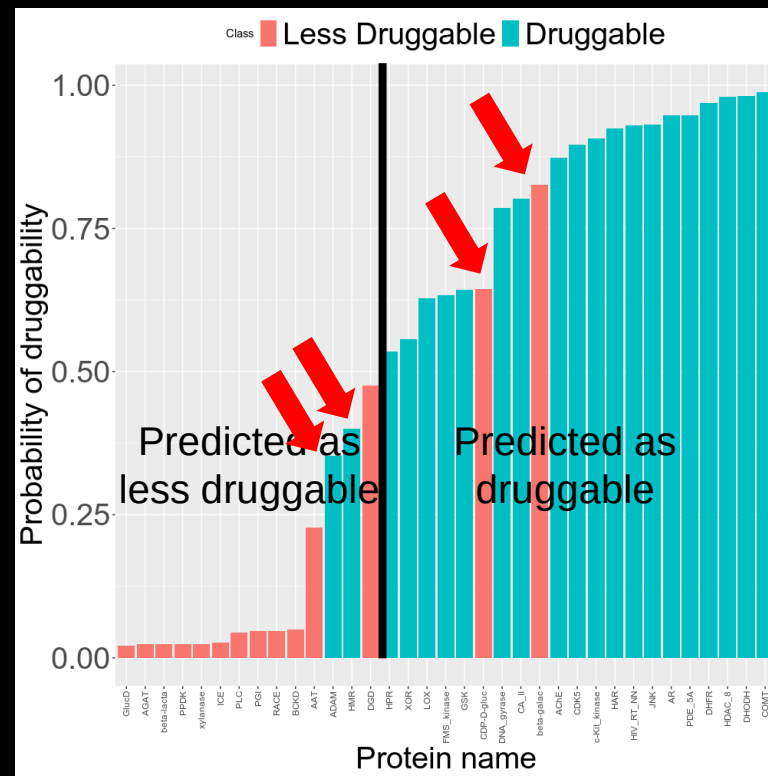


Protein training set

Training sets: 75 protein structures
99% callback accuracy



Validation set: 35 protein structures
89% callback accuracy

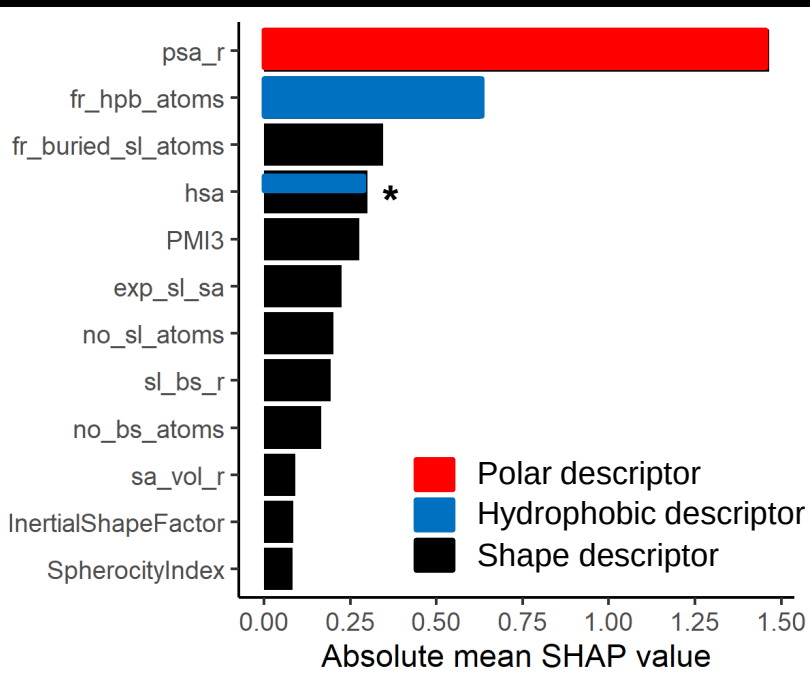


How do we know our model is behaving well?

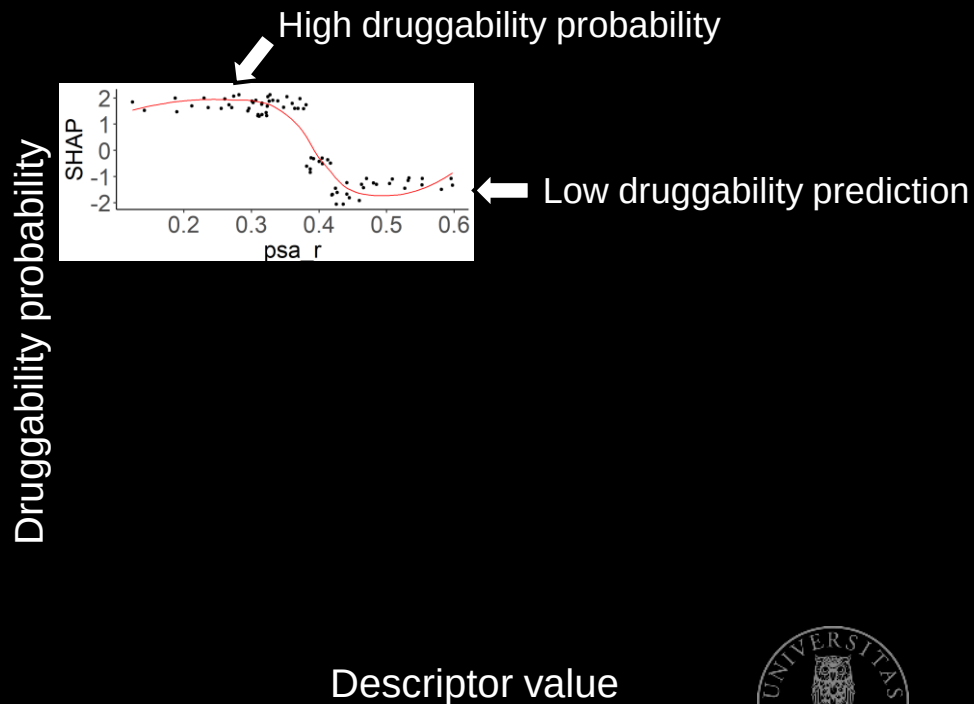


Which descriptors matter?

Introducing SHAP values

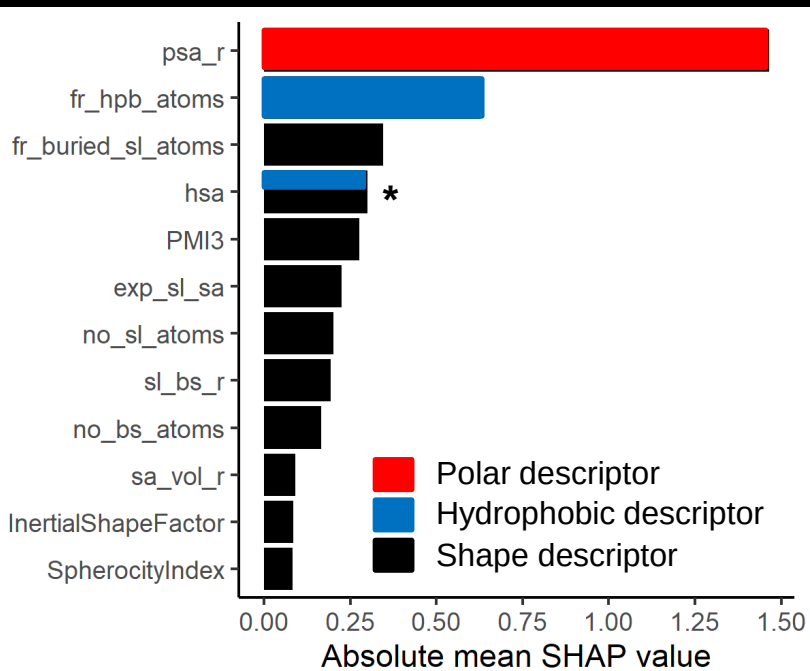


* Total hydrophobic surface area correlates with hydrophobicity AND general size

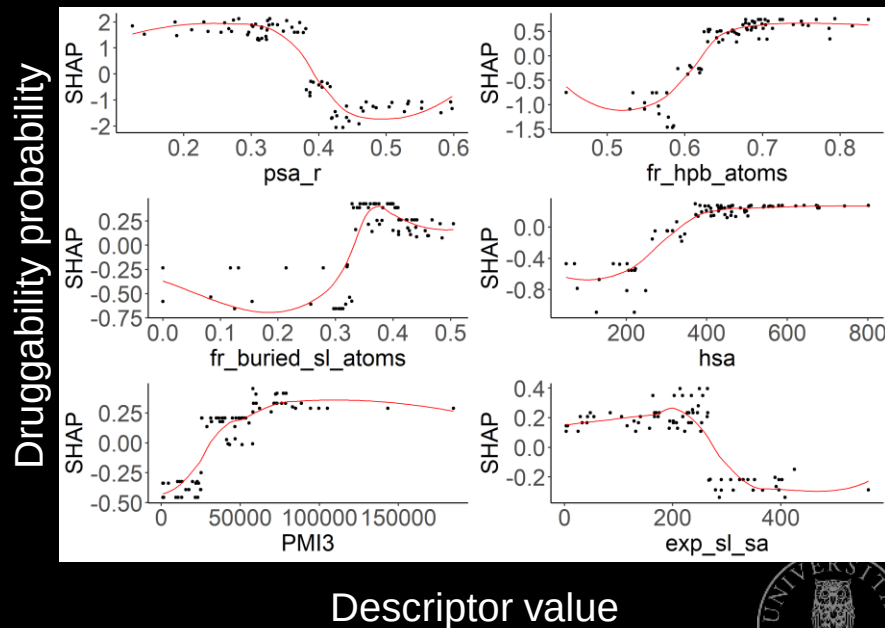


Which descriptors matter?

Introducing SHAP values



* Total hydrophobic surface area correlates with hydrophobicity AND general size

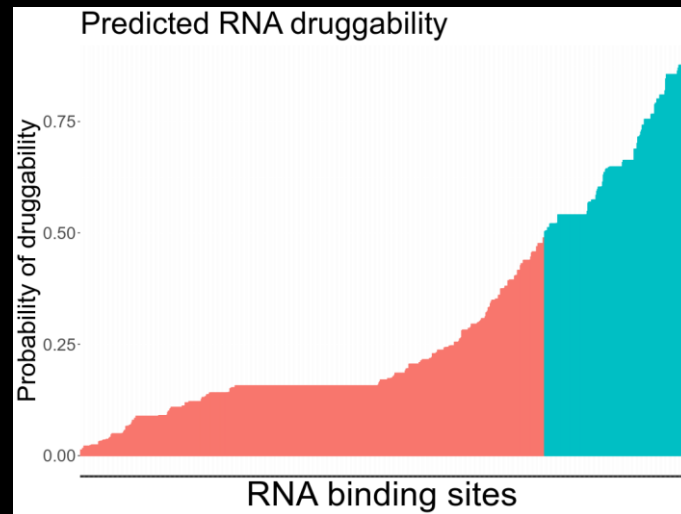


Then we moved on to RNA

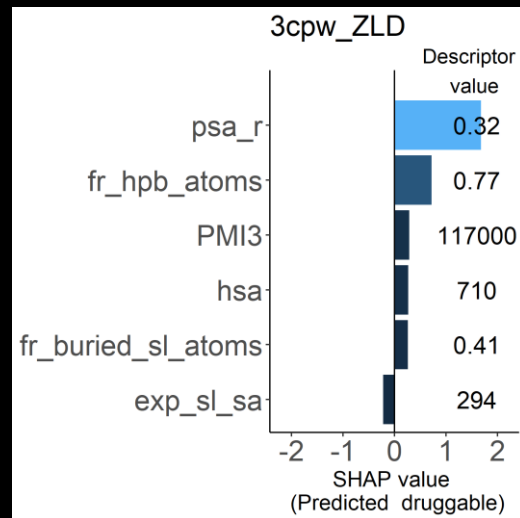
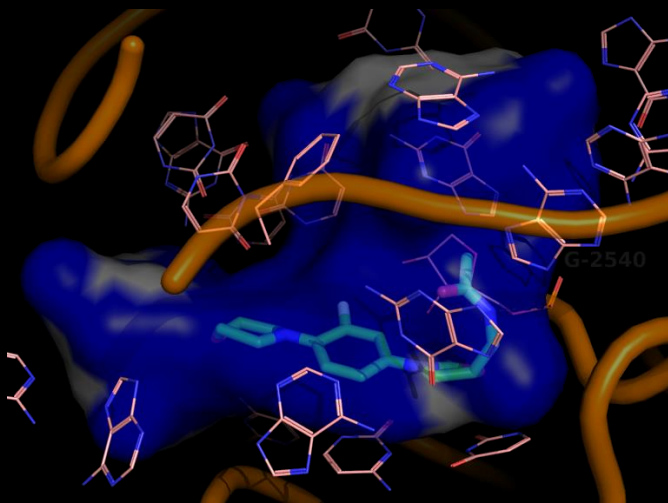


Creating our RNA data sets

	RNA-only set	Ribosome set
Unique PDB IDs	427	497
Binding sites containing small molecule ligands	465	613
Unique ligands	224	217
Druggable entries	172 (36%)	224 (31%)



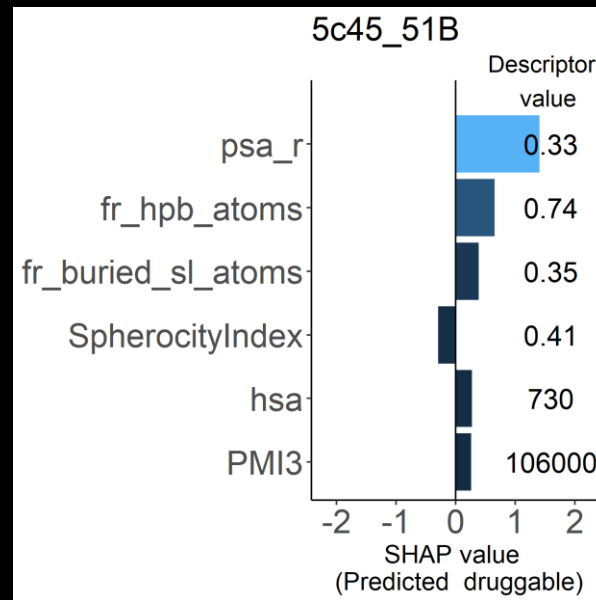
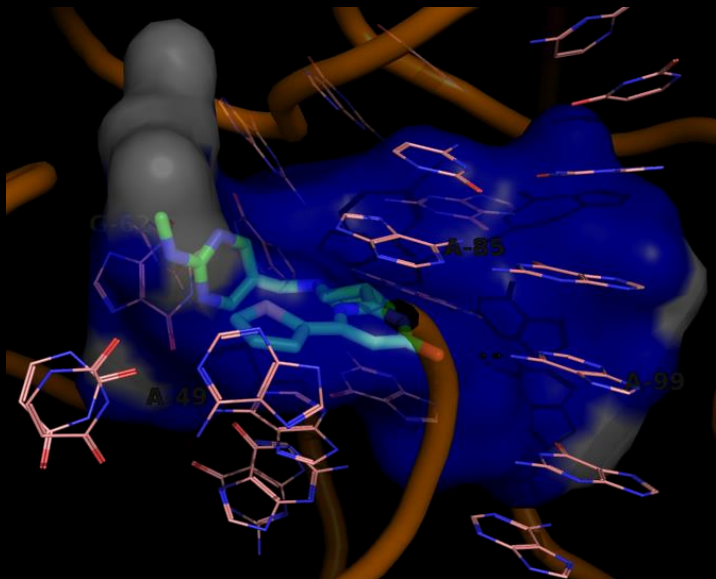
Evaluation of anecdotal examples: Linezolid, an FDA-approved drug



- Low relative polar surface area
- High abundance of hydrophobic atoms
- Beneficial shape



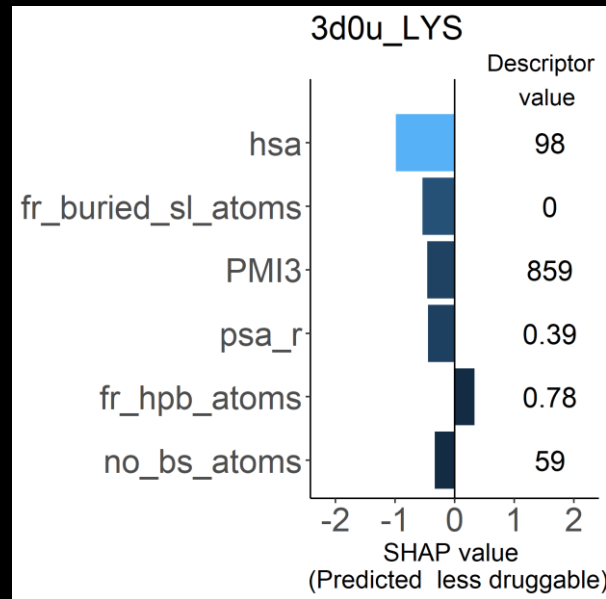
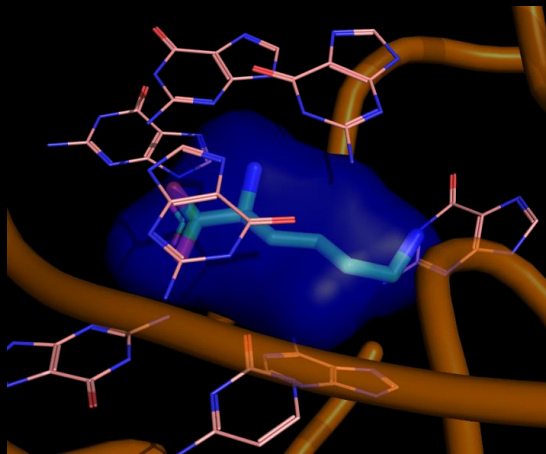
Evaluation of anecdotal examples: The FMN riboswitch, a promising antibiotic target



- Low relative polar surface area
- High abundance of hydrophobic atoms
- Large cavity



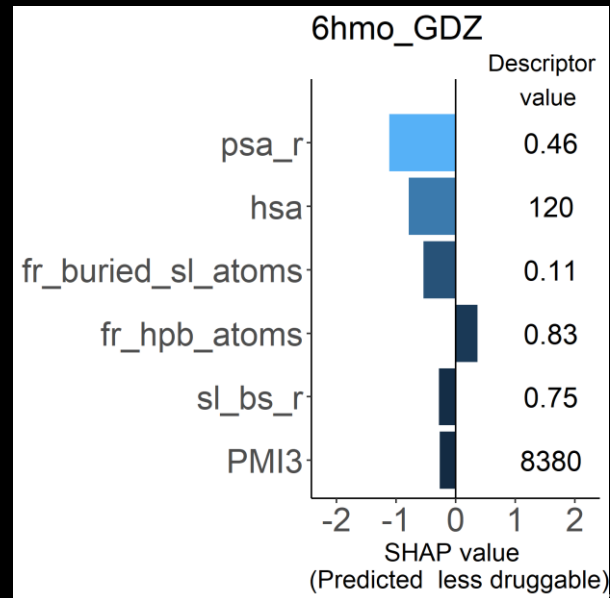
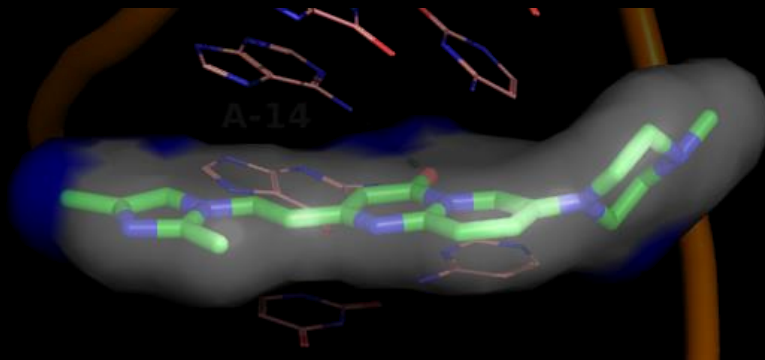
Evaluation of anecdotal examples: The lysine riboswitch, a less promising antibiotic target



- Small (hydrophobic) surface area
 - Small cavity
 - Lesser beneficial shape



Evaluation of anecdotal examples: A splicing site inhibitor



- Large relative polar surface area
- Small (hydrophobic) surface area
 - Flat shape



Drug-like ligands (QED ≥ 0.67) found in RNA binding sites predicted to be druggable.

Ligand ID	PDB ID	Receptor name	QED score	K_d [nM]	LE [kcal·mol ⁻¹ ·heavy atom ⁻¹]
RNA data set					
MGR	1q8n	Malachite green aptamer	0.76	800	0.34
6YG	5kx9	FMN riboswitch	0.69	13.4	0.41
L8H	2l8h	HIV-1 TAR RNA	0.67	NA	-
PMZ	1lvj	HIV-1 TAR RNA	0.85	27,000	0.22
Ribosomal data set					
917	5v7q	50S ribosomal subunit	0.94	700	0.39
ZLD	3cpw	50S ribosomal subunit	0.89	20,000	0.27
G6M	6ddg	50S ribosomal subunit	0.79	2,600	0.31
3HE	4u3u	80S ribosome	0.76	140	0.48
G6V	6ddd	50S ribosomal subunit	0.76	2,600	0.30
ANM	3cc4	50S ribosomal subunit	0.78	20,000	0.34
HN8	5on6	80S ribosome	0.71	NA	-
3K8	4u55	80S ribosome	0.71	39	0.32

- Most druggable ligands bind to druggable binding sites
- Mid-to-high affinities (K_D)
- Good ligand efficiencies (LE > 0.30 kcal·mol⁻¹·heavy atom⁻¹)



Drug-like ligands ($\text{QED} \geq 0.67$) found in RNA binding sites

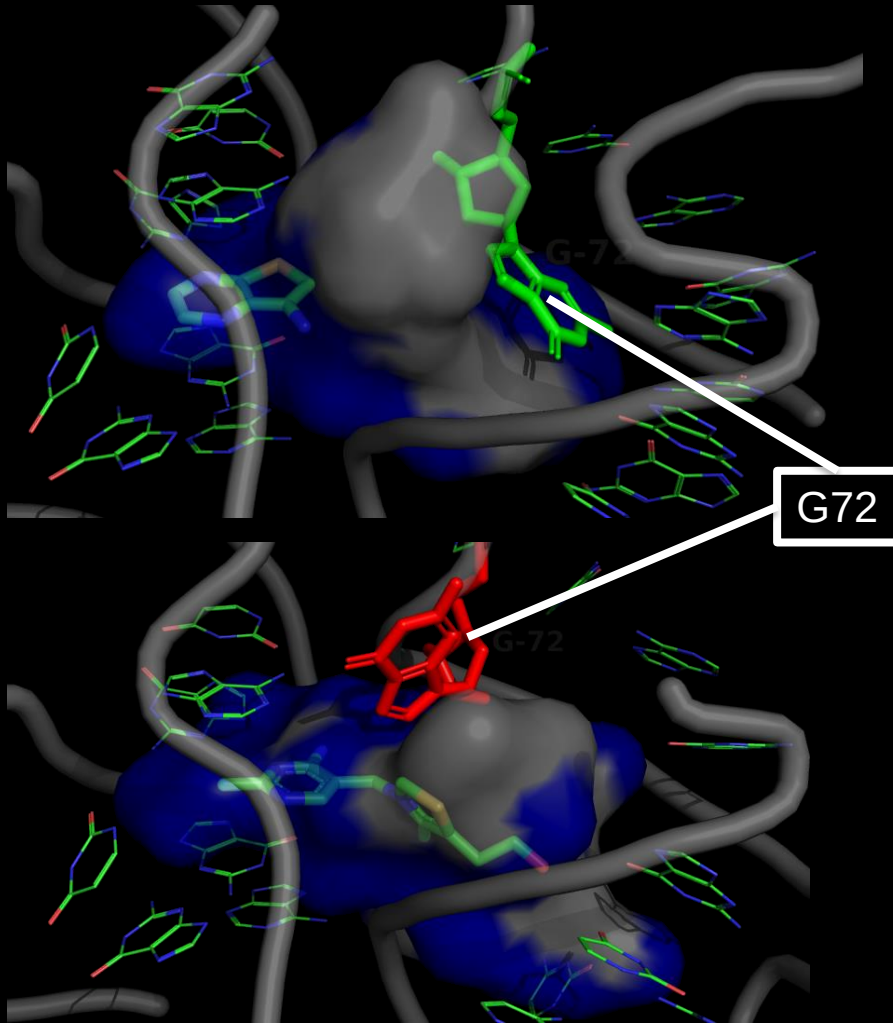
predicted to be less druggable.

Ligand ID	PDB code	Receptor name	QED	K_D [nM]	LE [kcal·mol ⁻¹ ·heavy atom ⁻¹]
RNA data set					
VIB	4nyg	TPP riboswitch	0.79	1,500	0.45
2QC	4nyb	TPP riboswitch	0.77	103,000	0.43
0EC	2lwk	Influenza A	0.86	50,000	0.29
1TU	5ob3	Spinach aptamer	0.85	530	0.49
218	2hop	TPP riboswitch	0.77	6,000	0.38
Ribosomal data set					
	4v6o	Tryptophan- sensing ribosomal site		NA#	-
TRP			0.67		

- Fewer drug-like ligands in lesser druggable sites
- Moderate affinities (K_D)
- Misclassification

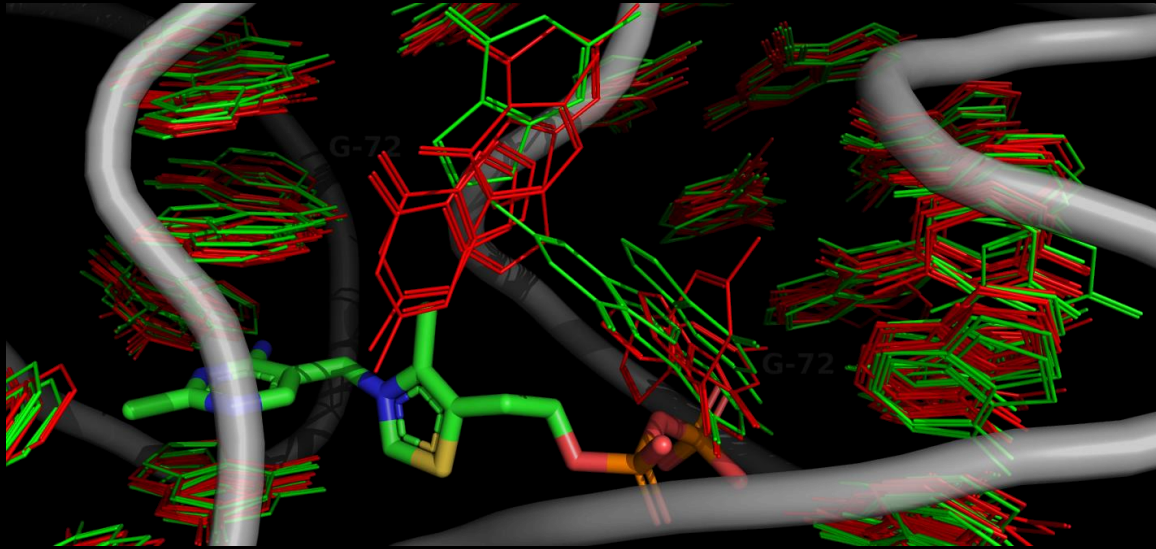


Conformations can affect the druggability prediction



- The residue G72 in the TPP riboswitch can impact the size and the shape of the binding site
- Different predictions for same binding site:
 - Green: druggable
 - Red: less druggable

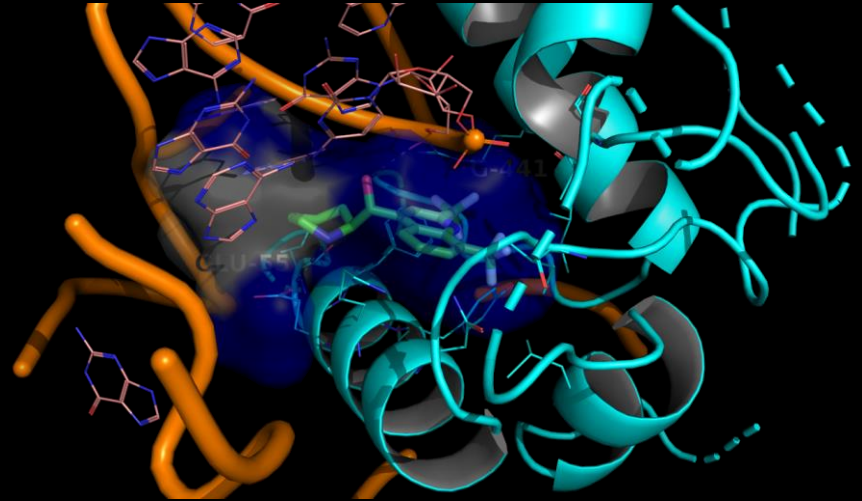
Conformations can affect the druggability prediction



- The residue G72 can impact the size and the shape of the binding site
- Green: druggable
- Red: less druggable

Conclusion

- Successfully created a robust druggability prediction method
- Compliant with RNA and protein
- High accuracy
- Model behaviour on both protein testing set and RNA sample set as expected



RNA as a promising target for future antibiotics



The antibiotic age

- From 1900- 2000:
Infectious-related deaths nearly annihilated
- Improved life span and quality of life partially attributed to antibiotics

Figure deleted as it cannot be spread under an open license

Figure 1 from the citation below

Armstrong, G. L.; Conn, L. A.; Pinner, R. W. Trends in Infectious Disease Mortality in the United States During the 20th Century. *JAMA* **1999**, *281* (1), 61–66.
<https://doi.org/10.1001/jama.281.1.61>.

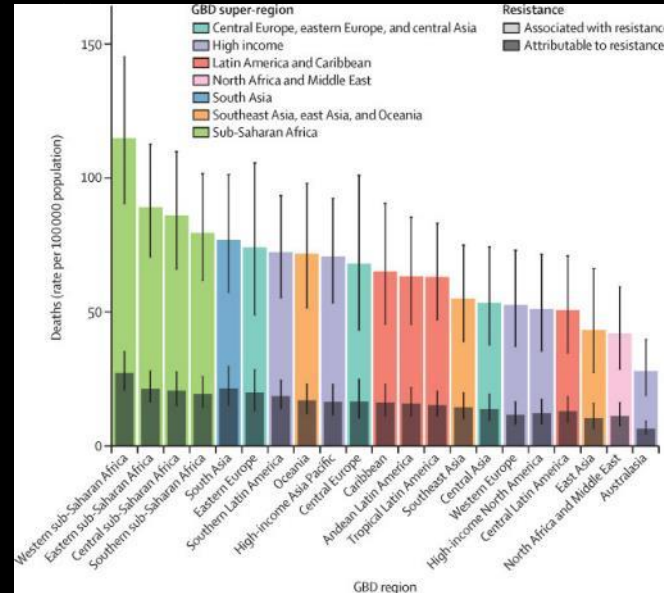
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The increasing problem of antimicrobial resistance (AMR)

- Infectious-related deaths are becoming more common
- Class-wide resistance

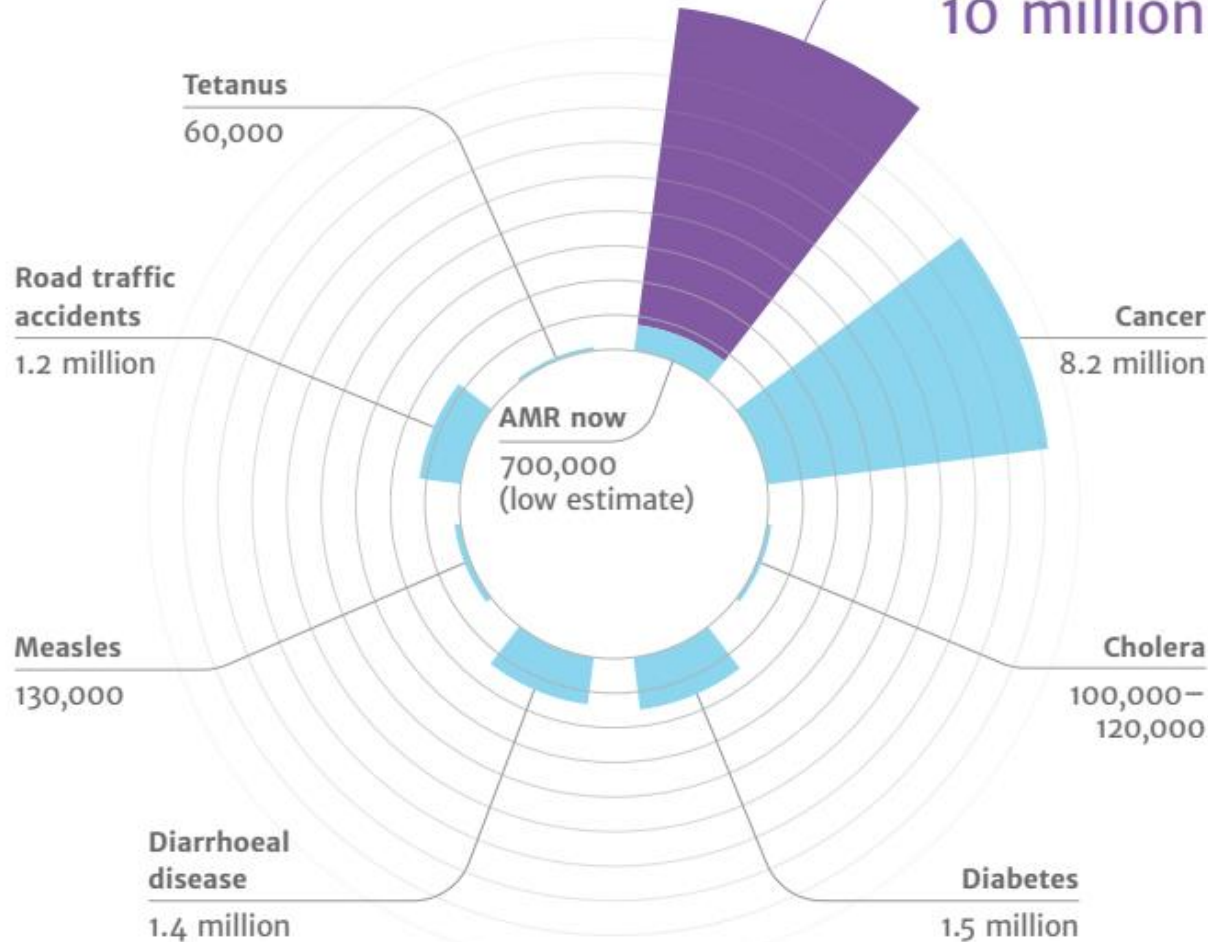
All-age rate of deaths attributable to and associated with bacterial antimicrobial resistance by GBD region,



Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis
Lancet, 2019 [https://doi.org/10.1016/S0140-6736\(21\)02724-0](https://doi.org/10.1016/S0140-6736(21)02724-0)
CC BY-4.0



AMR in 2050 10 million

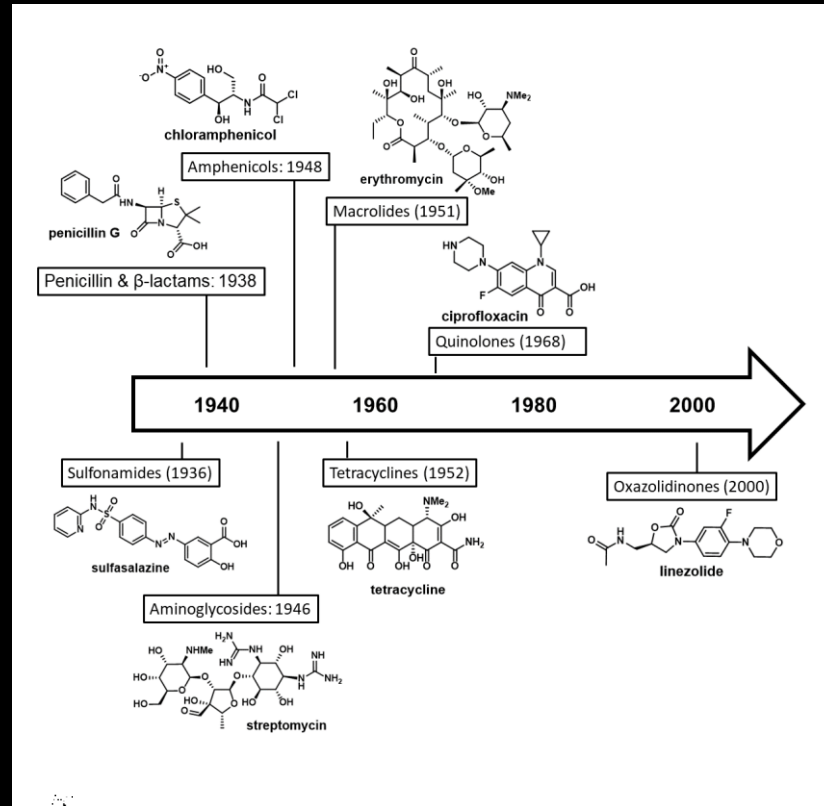


O'Neill, J. *Antimicrobial Resistance: Tackling a Crisis for the Health and Wealth of Nations*; Wellcome Collection, 2014.
CC-BY-4.0



Timeline of antibiotic research

- «Golden Age»: 1950s-1960s
- To date: 210 FDA approved antibiotics
- Since 2000s: only one new class of antibiotics approved



Antibacterial resistance

- Bacteria are able to adapt and overcome antibiotic susceptibility
- Not a new phenomena
- WHO projected by 2050 to cause more deaths than cancer

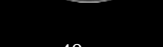
“It is not difficult to make microbes resistant to penicillin in the laboratory by exposing them to concentrations not sufficient to kill them [...] Then there is the danger that the ignorant man may easily underdose himself and by exposing his microbes to non-lethal quantities of the drug make them resistant.”

– Alexander Fleming, Nobel acceptance speech, 1945



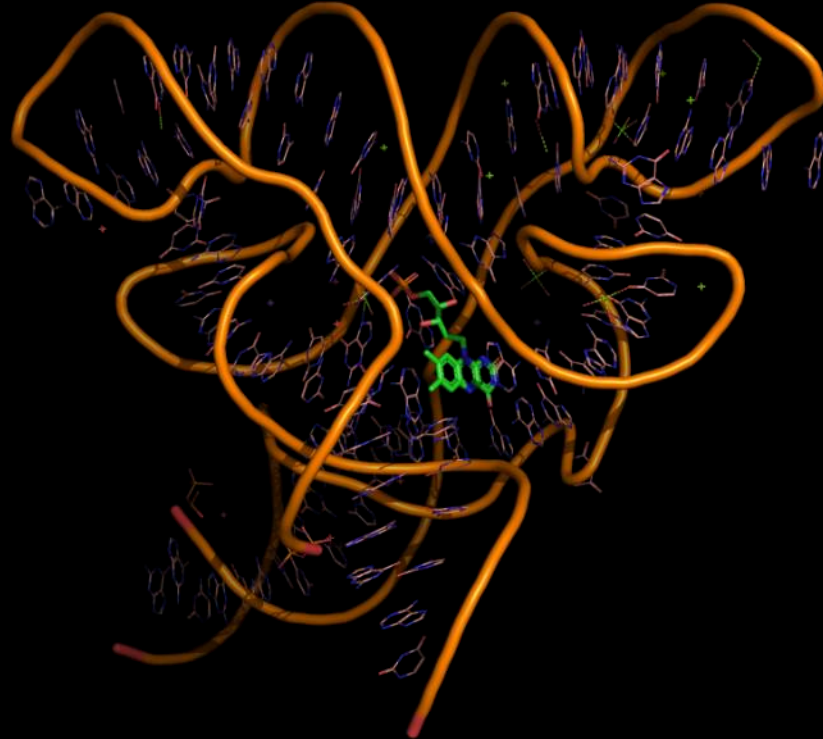
**Riboswitches:
a novel target for a potential new
class of antibiotics**



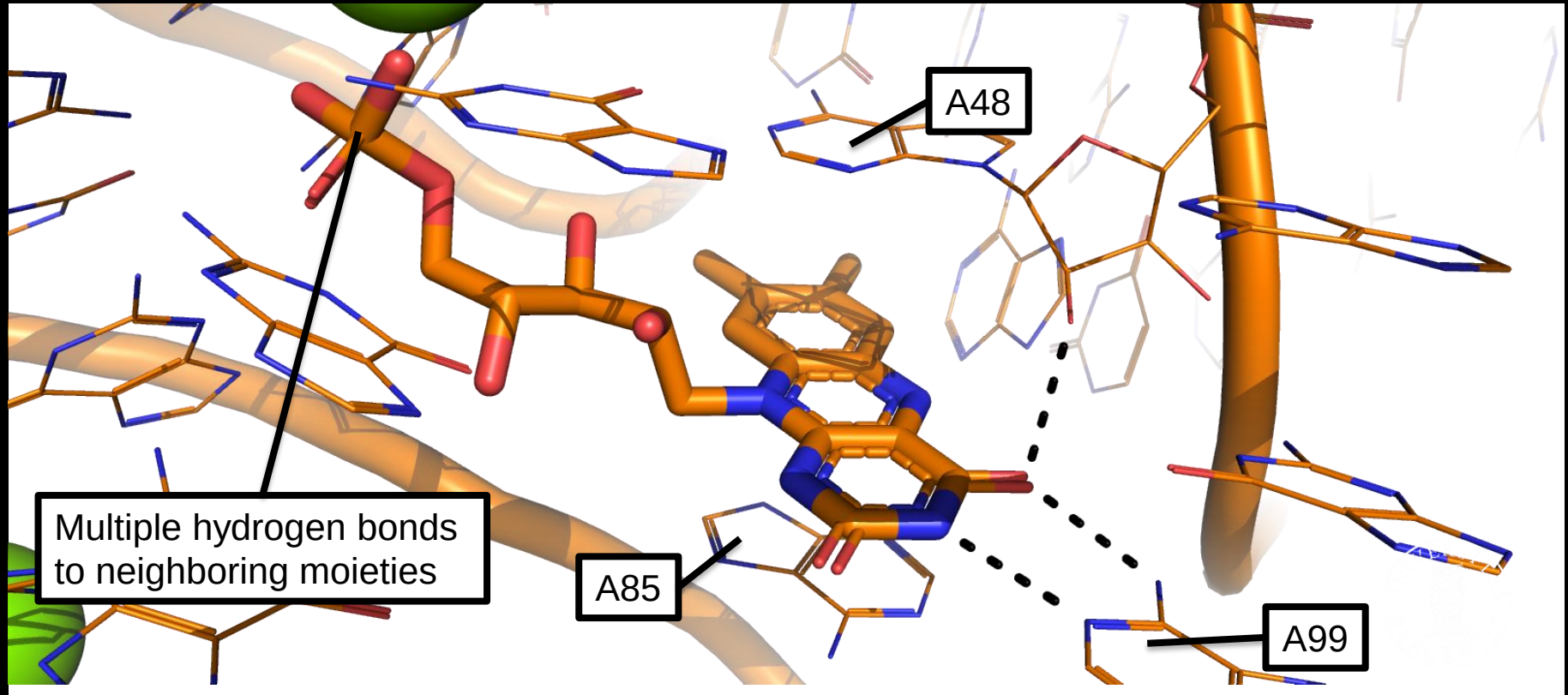
NC(CCCC(=O)O)C

The FMN riboswitch

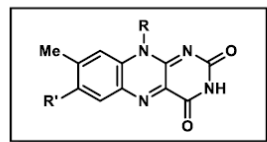
- Binds to riboflavin mononucleotide (FMN)
- Regulates riboflavin biosynthesis
- Abundant in many pathogens
- Large binding site
- Multiple known ligands



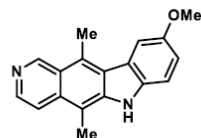
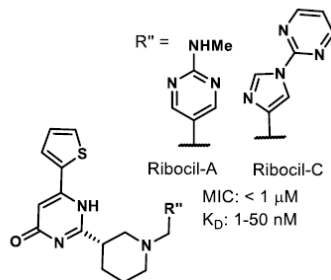
Binding mode of FMN



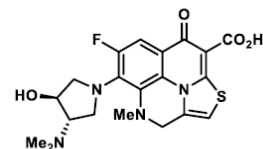
Known ligands



Riboflavin scaffold



WG-1
MIC: 3.1 μ M
K_D: ND



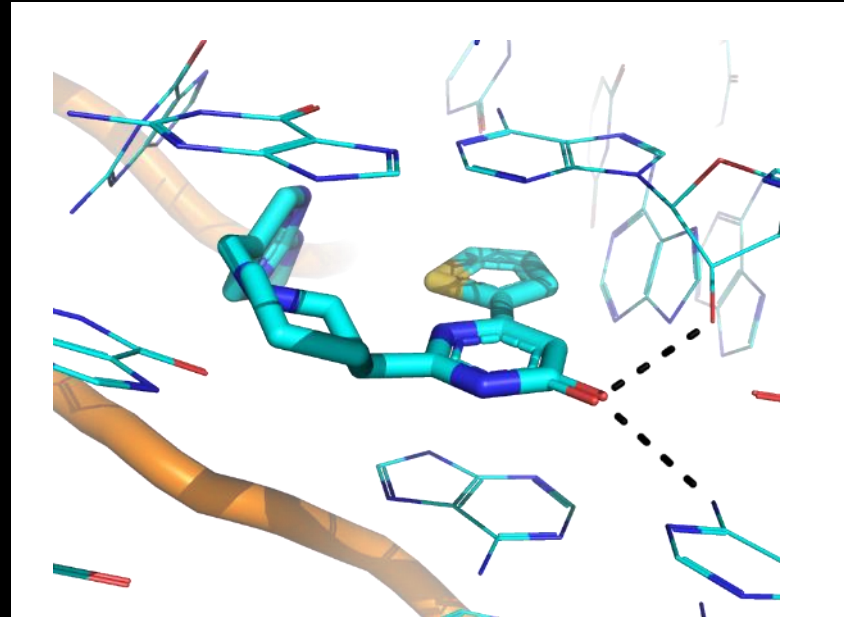
WG-3
MIC: <0.1 μ M
K_D: ~130 nM

Compound	R =	R' =	K _D	Compound	R =	R' =	K _D
Riboflavin			~39 μ M	Roseoflavin (RoF)			~0.2-1 μ M
FMN			~1-37 nM	RoFMN			~3 nM
BRX830			~2.6 nM	5FDQD			~7.5 nM
BRX1555			~39 nM	BRX1354			~16 nM



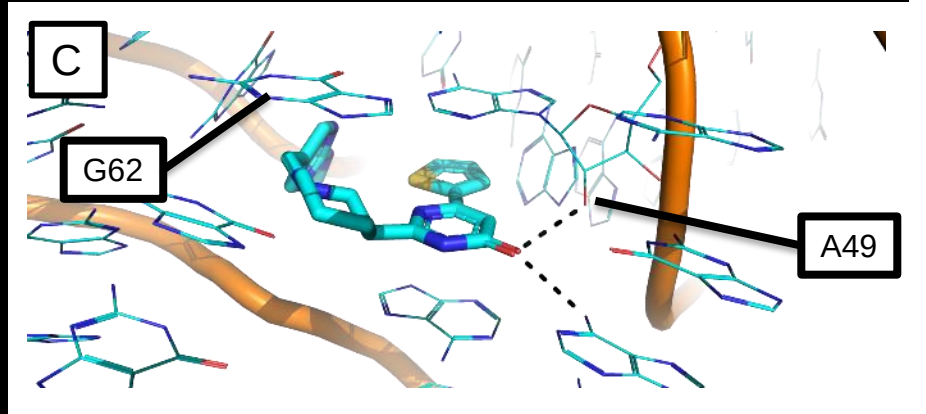
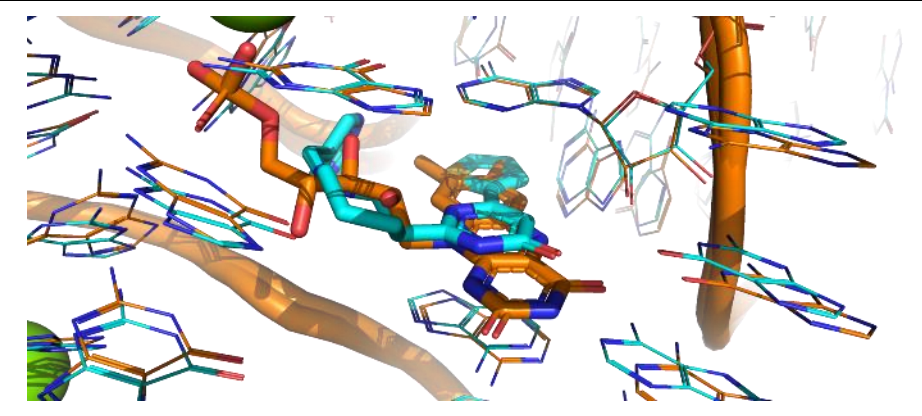
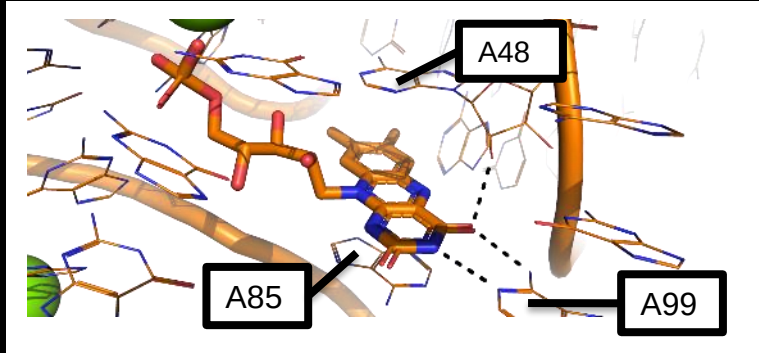
Ribocil: a promising hit

- Discovered by Merck
- Highly drug-like molecule
- Highly potent binder
- Structurally different from FMN
- Some overlapping binding modes

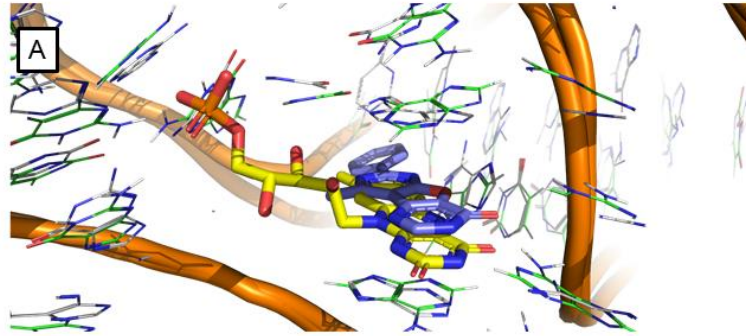


Ribocil

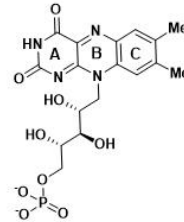
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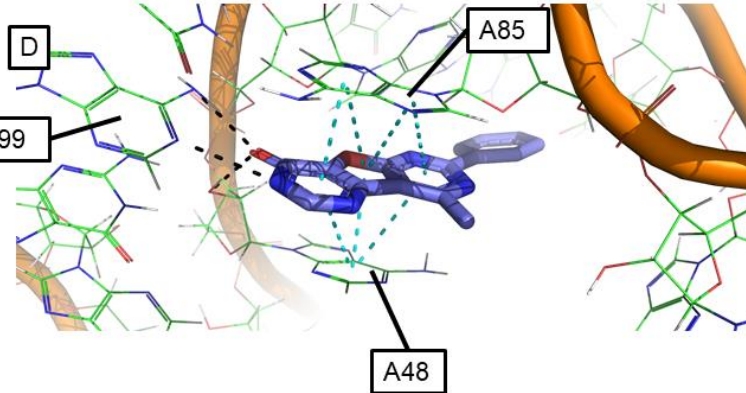
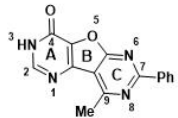
RSL-0035: A previous hit in the Brenk group



B



C



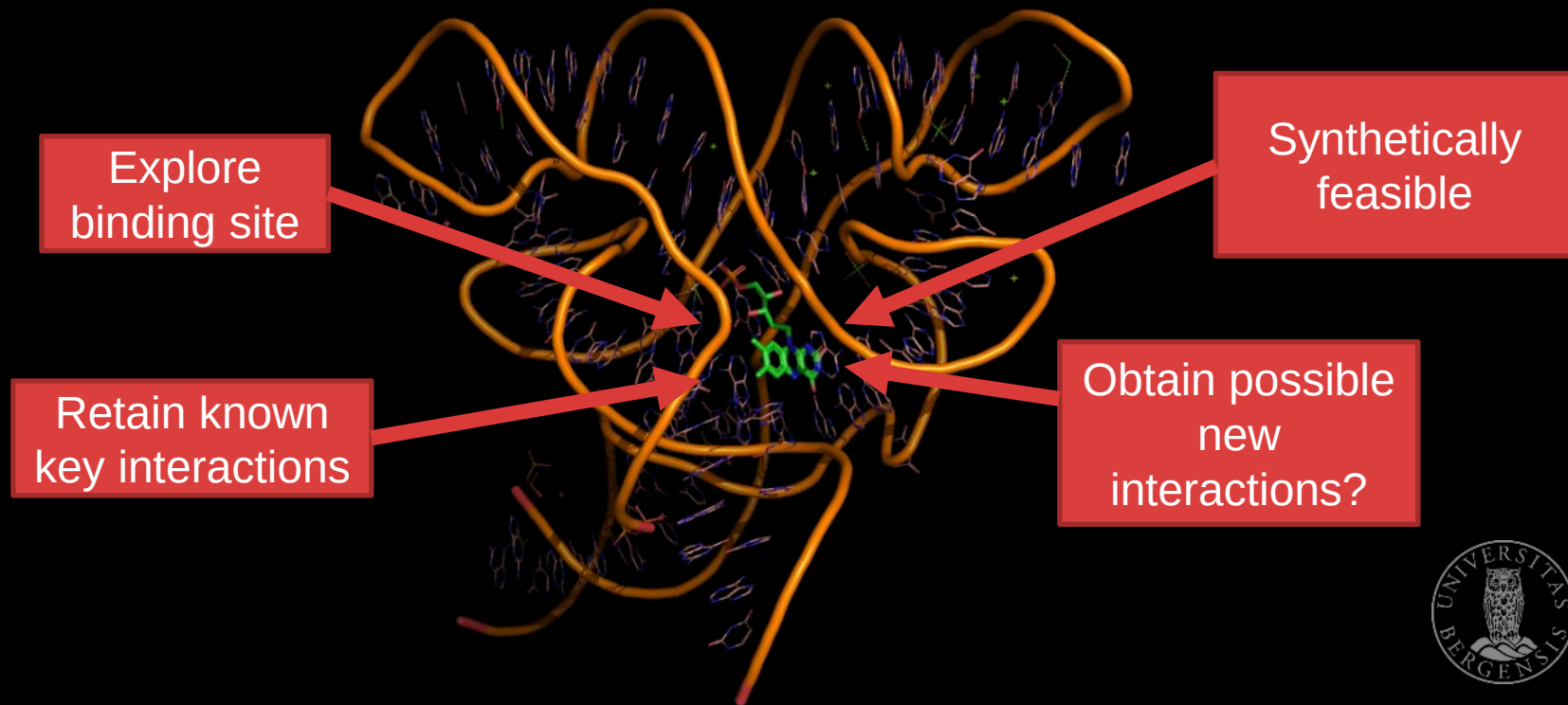
- Virtual screening hit
- Similar predicted binding mode



Ligand design

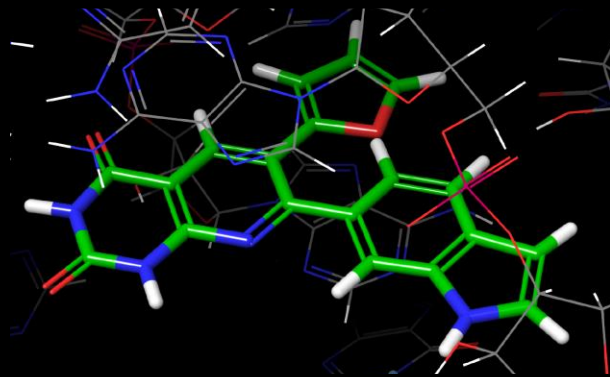


What new ligands can we propose for the FMN riboswitch?



Guidelines for ligand design

- Favor hydrophobic interactions: aromatic π/π stacking or edge/face interactions
- Seek out hydrogen bonding if possible
- Aim for efficiency (high LE); avoid extensive bridging

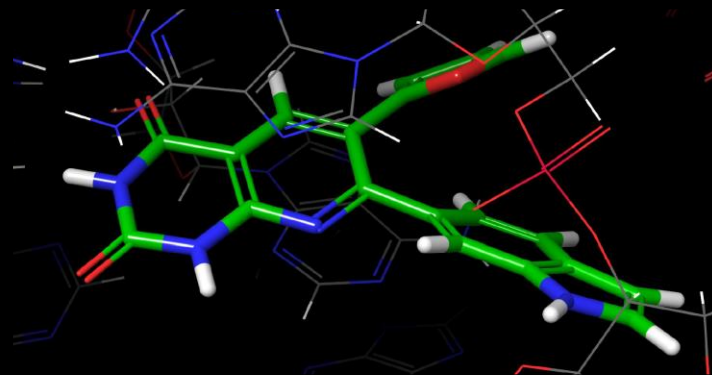


$$\text{Ligand Efficiency (LE)} = \frac{\text{Binding energy}}{N_{\text{Heavy atoms}}}$$



Methodology

- Rational approach
- Build structures “in silico”
- Minimize ligand/binding site complex
- Assess predicted binding mode

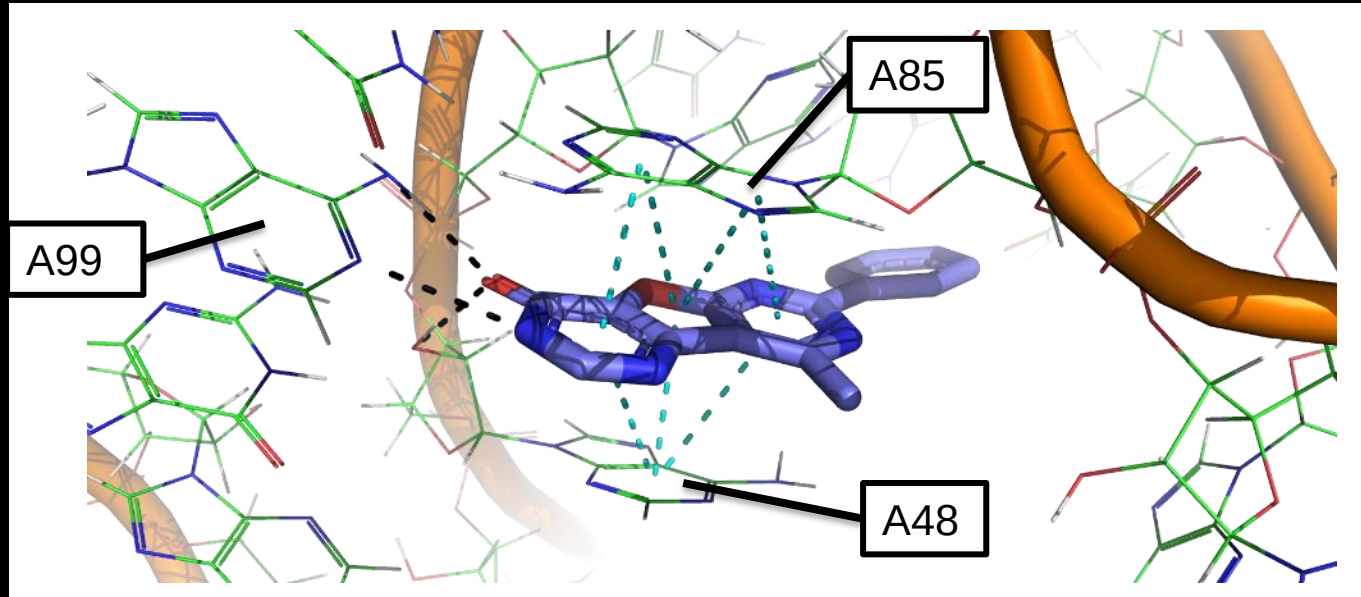


- Atom coordinates
- Bond angles, distances
- Atom distances: Coulomb, vdW

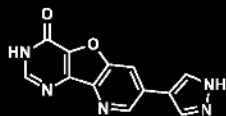


Criteria for inspection

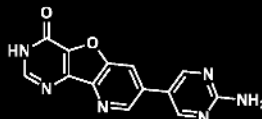
- Retain sandwiching
- Retain hydrogen bonds



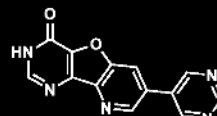
New proposed ligands for the FMN RS



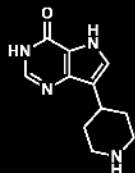
RSL-0029



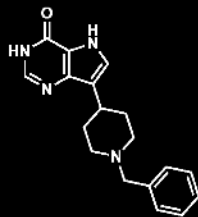
RSL-0026



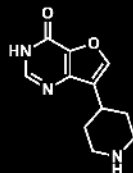
RSL-0027



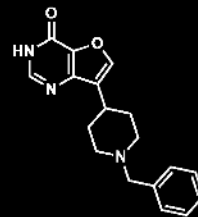
RSL-0012



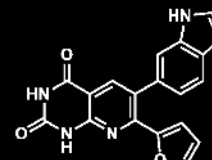
RSL-0007



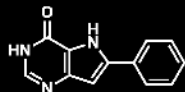
RSL-0015



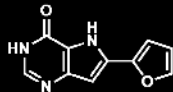
RSL-0014



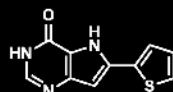
RSL-0036



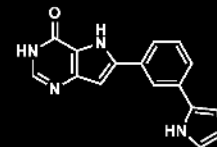
RSL-0031



RSL-0032



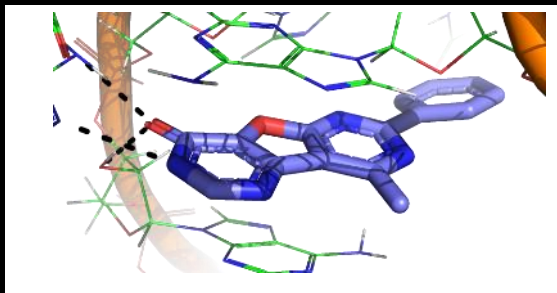
RSL-0033



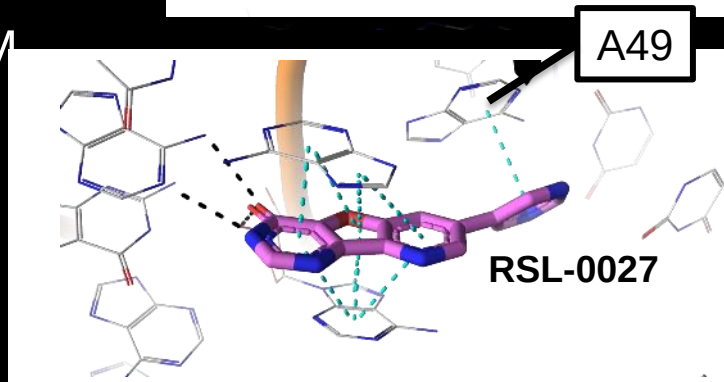
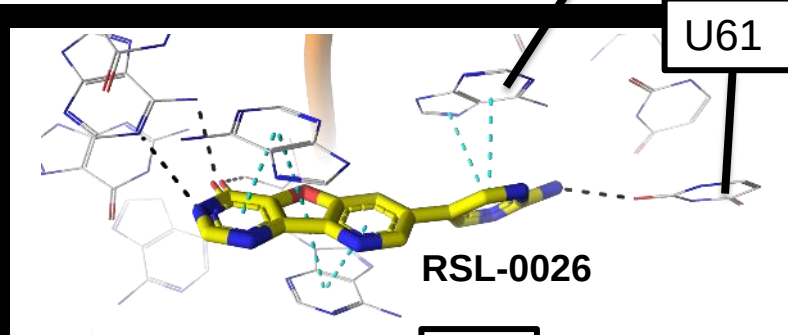
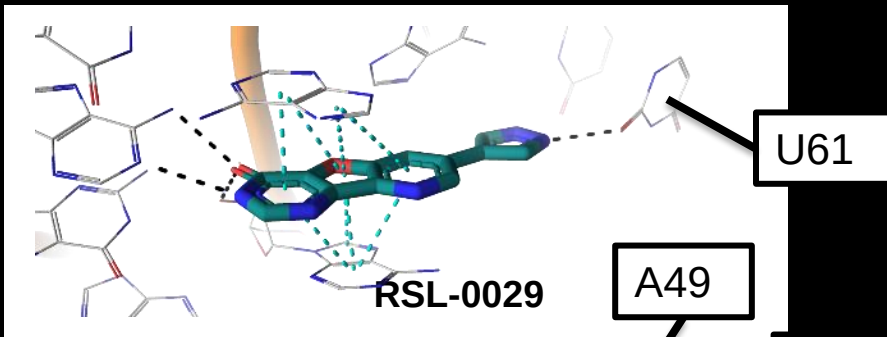
RSL-0034

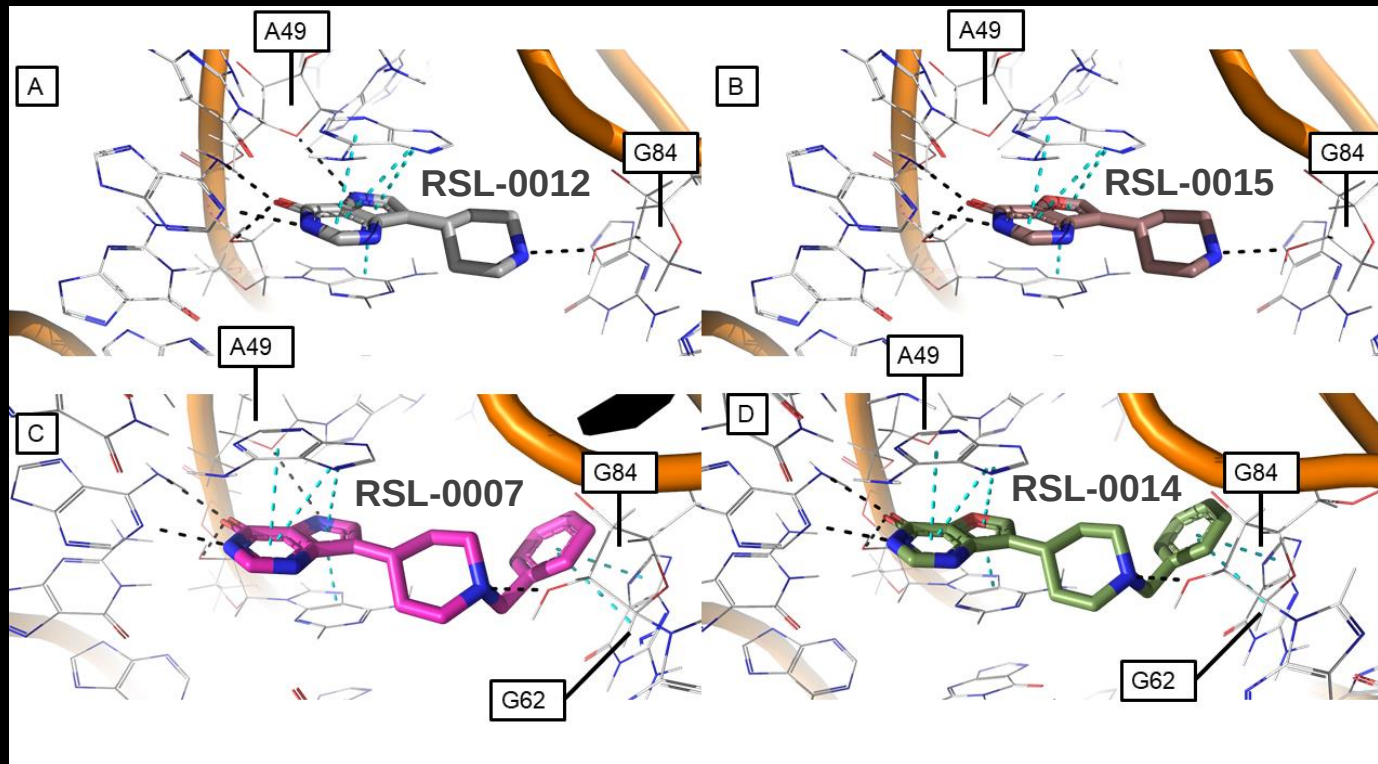


Inspired by RSL-0035



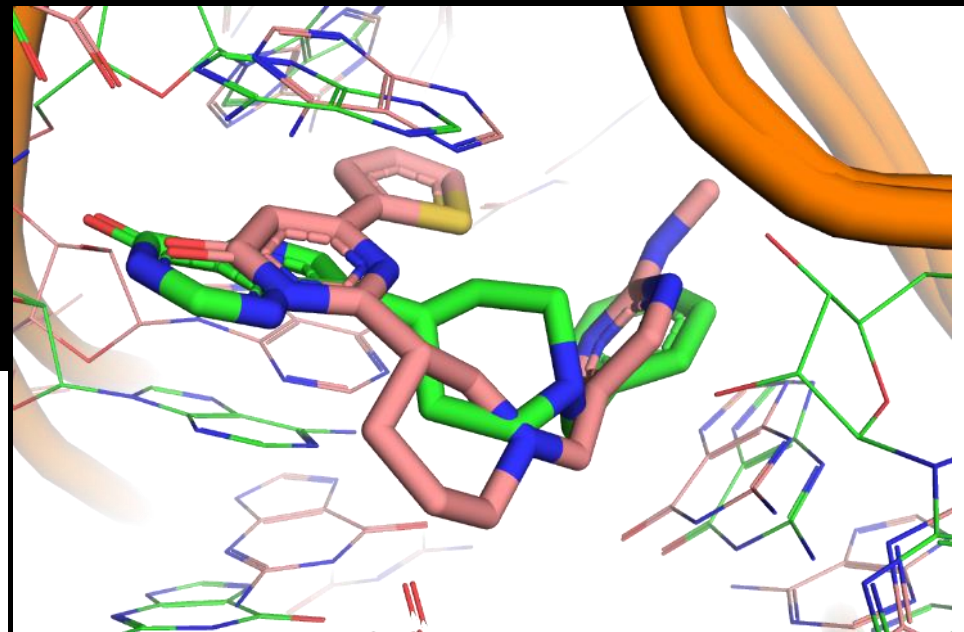
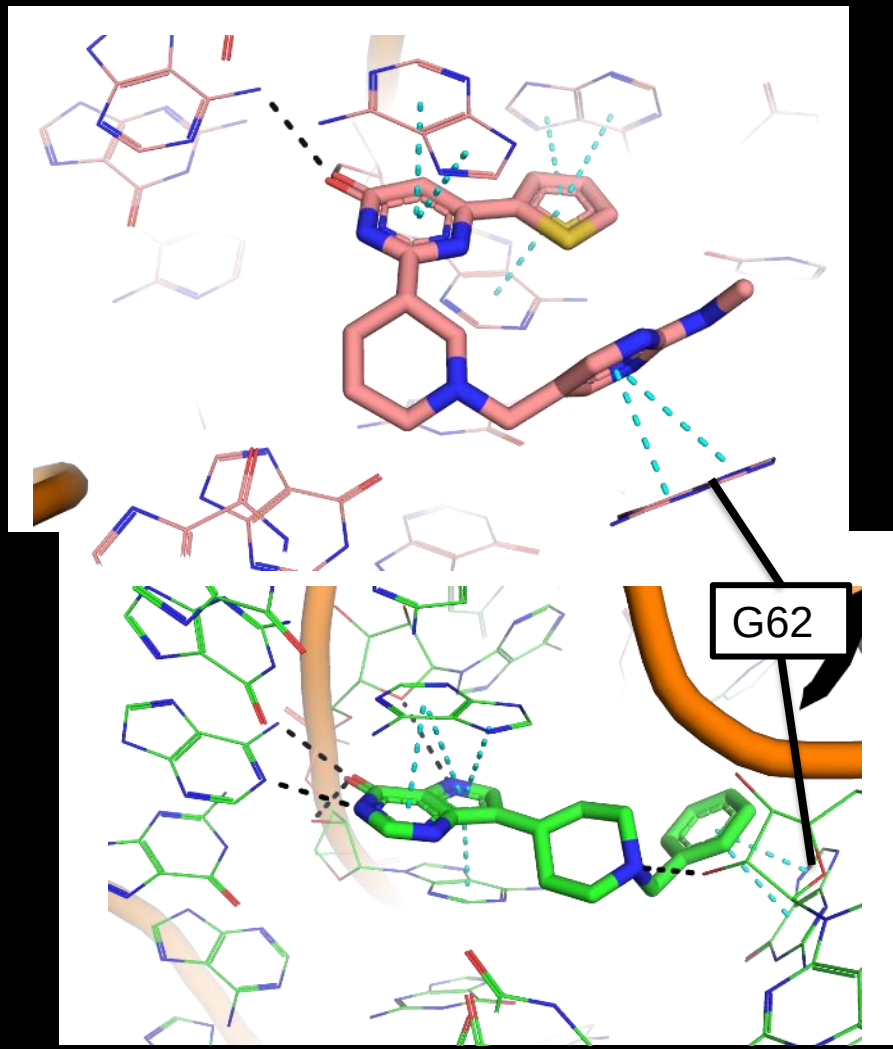
- Predicted binding modes:
New hydrophobic, polar interactions
- Synthesized by Valeriia Burova, M.Sc. & Dr. Muhammed Zeeshan

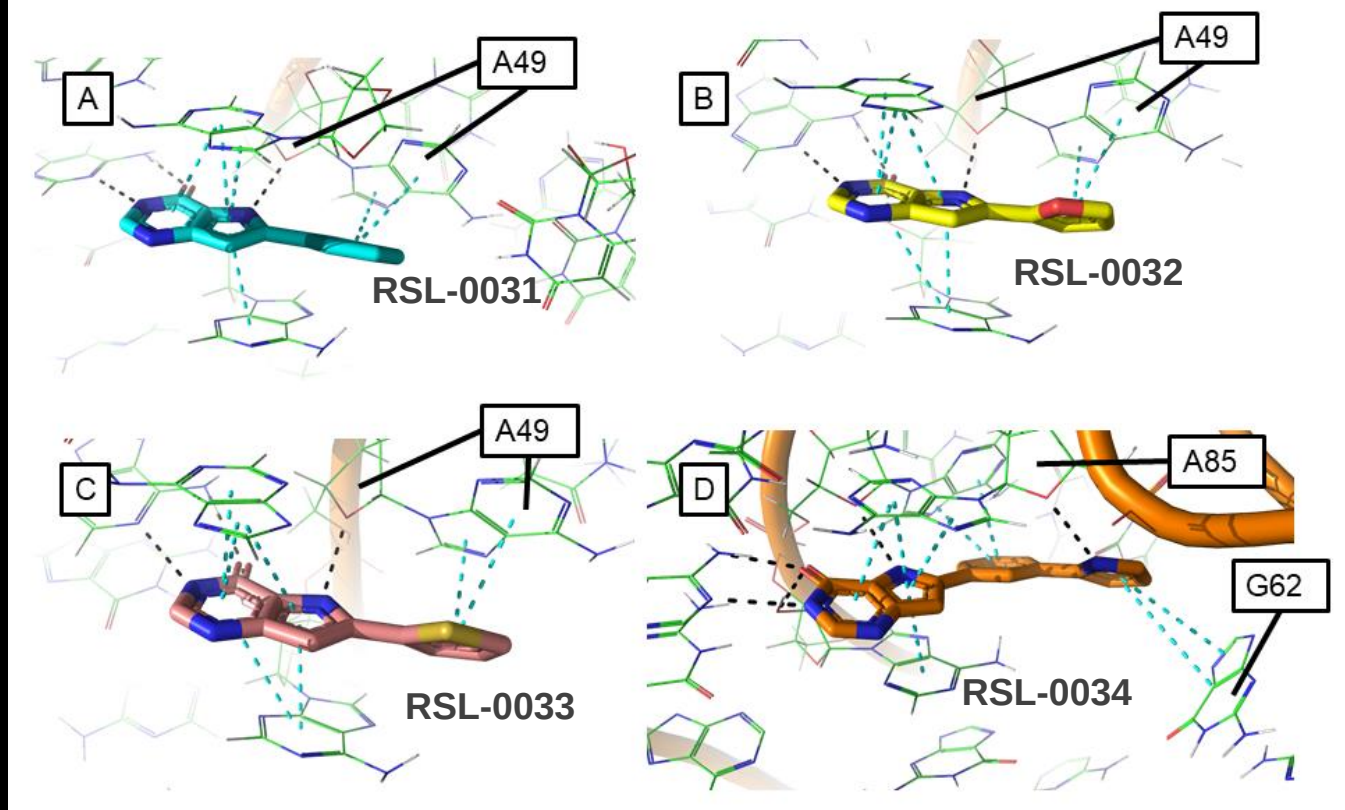




- Smaller scaffold, new vector
- Potential hydrogen bond (A49, G84)
- Potential additional stacking (G62) (Mimics ribocil)
- Synthesis by Dr. Muhammed Zeeshan

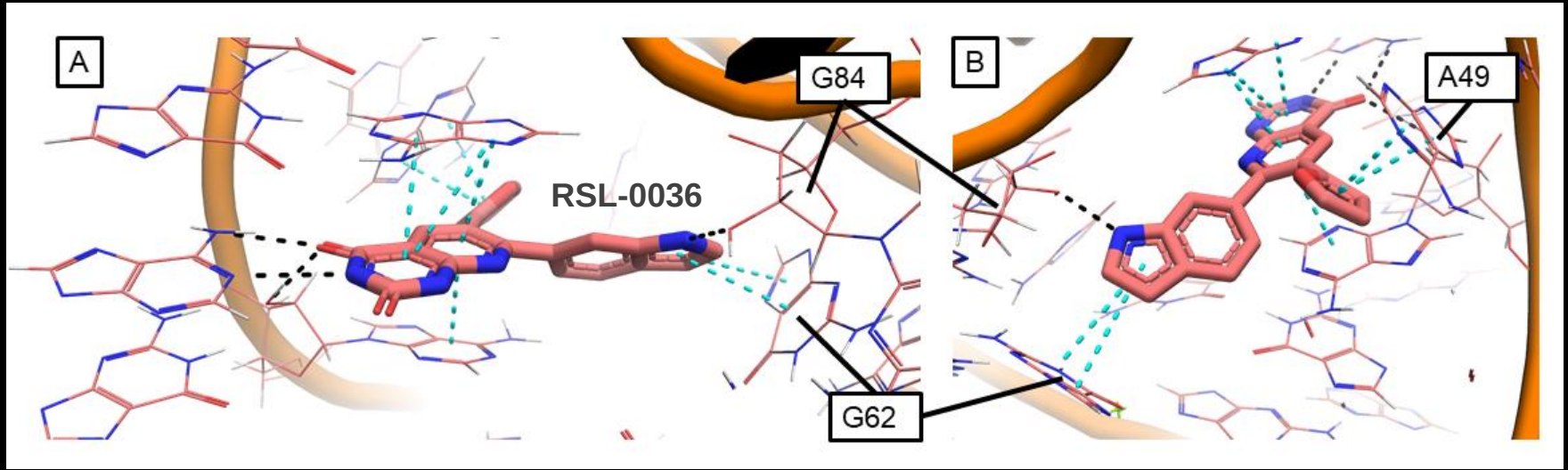






- Same scaffold, new vector
- New π/π -interaction (A49)
- Different aromatic substituents
- Possibility for branching to G62
- Synthesis shown later





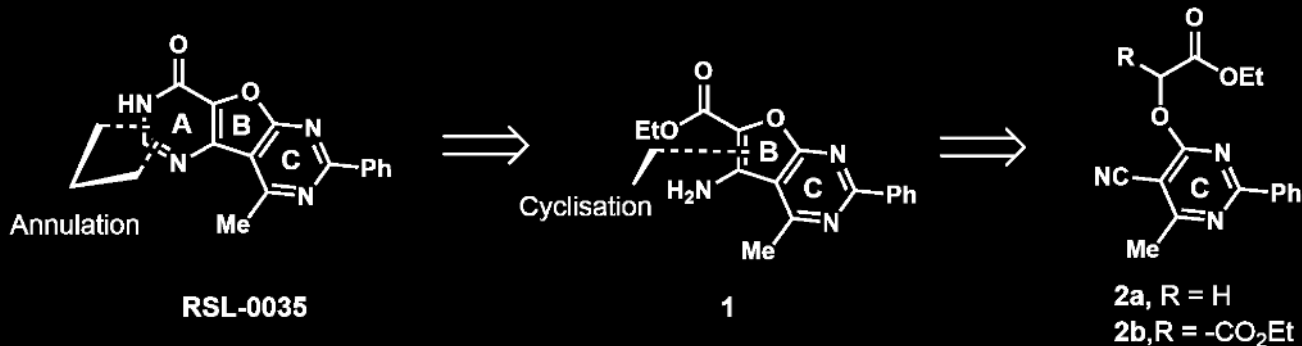
- New scaffold
- Possible to explore two vectors



Synthesis



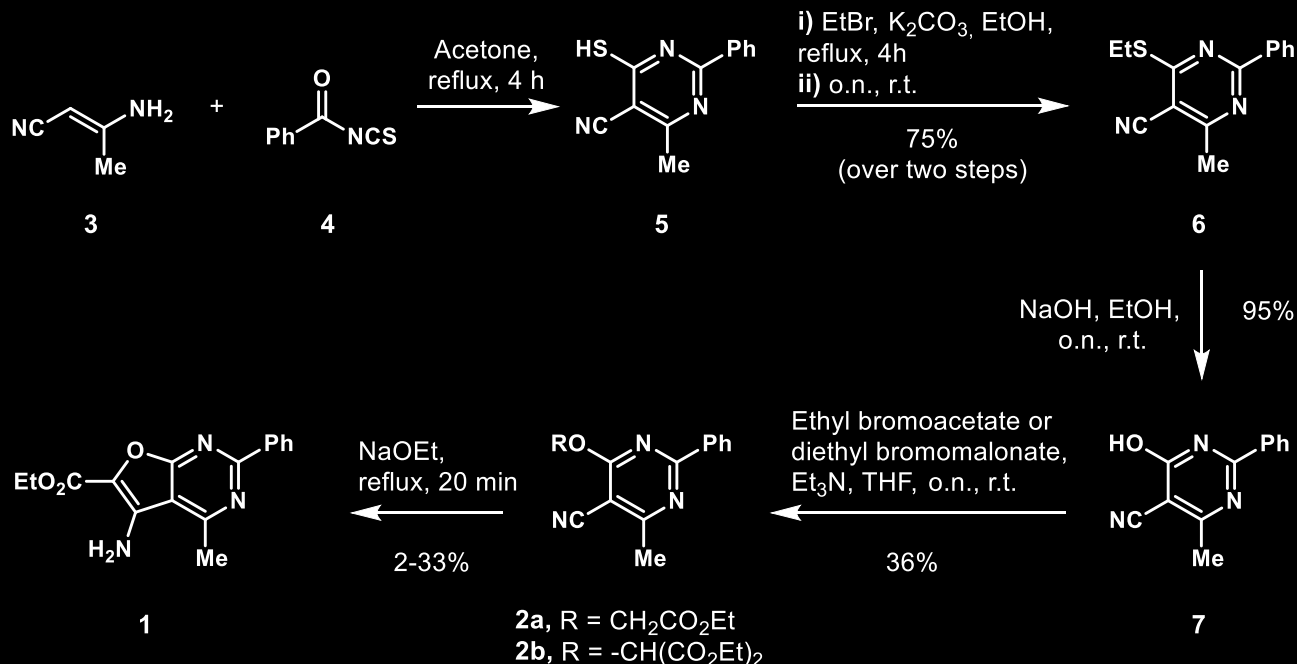
RSL-0035, a previous hit



- Stepwise construction of rings in scaffold
- Annulation, cyclisation
- Synthesis previously published



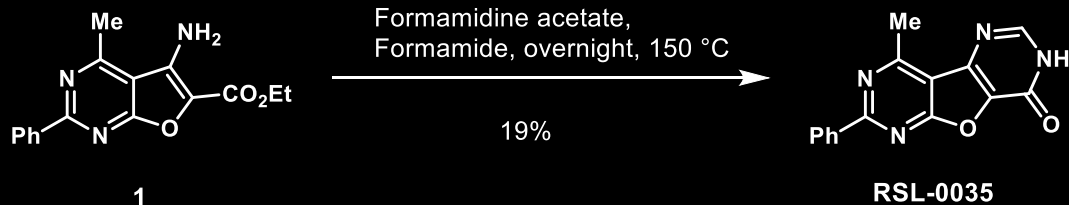
Synthesis of RSL-0035 I



- Optimization of **2a/2b** → **1**



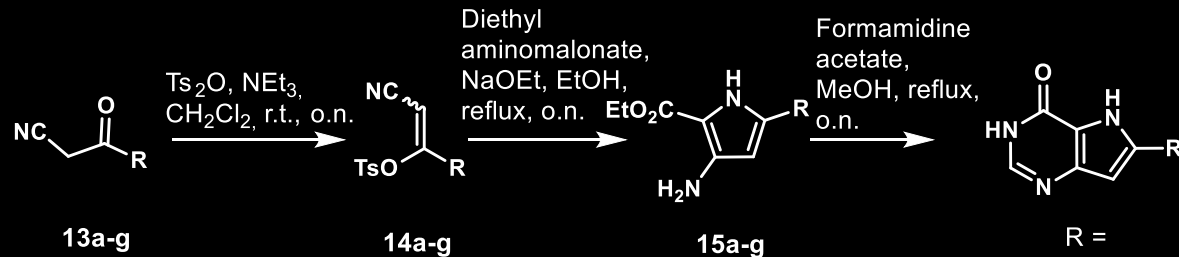
Synthesis of RSL-0035 II



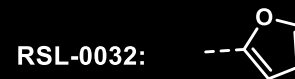
- Obtained after 5 steps
- Purified
- Stability issues



Synthesis of RSL-0031, RSL-0032, RSL-0033



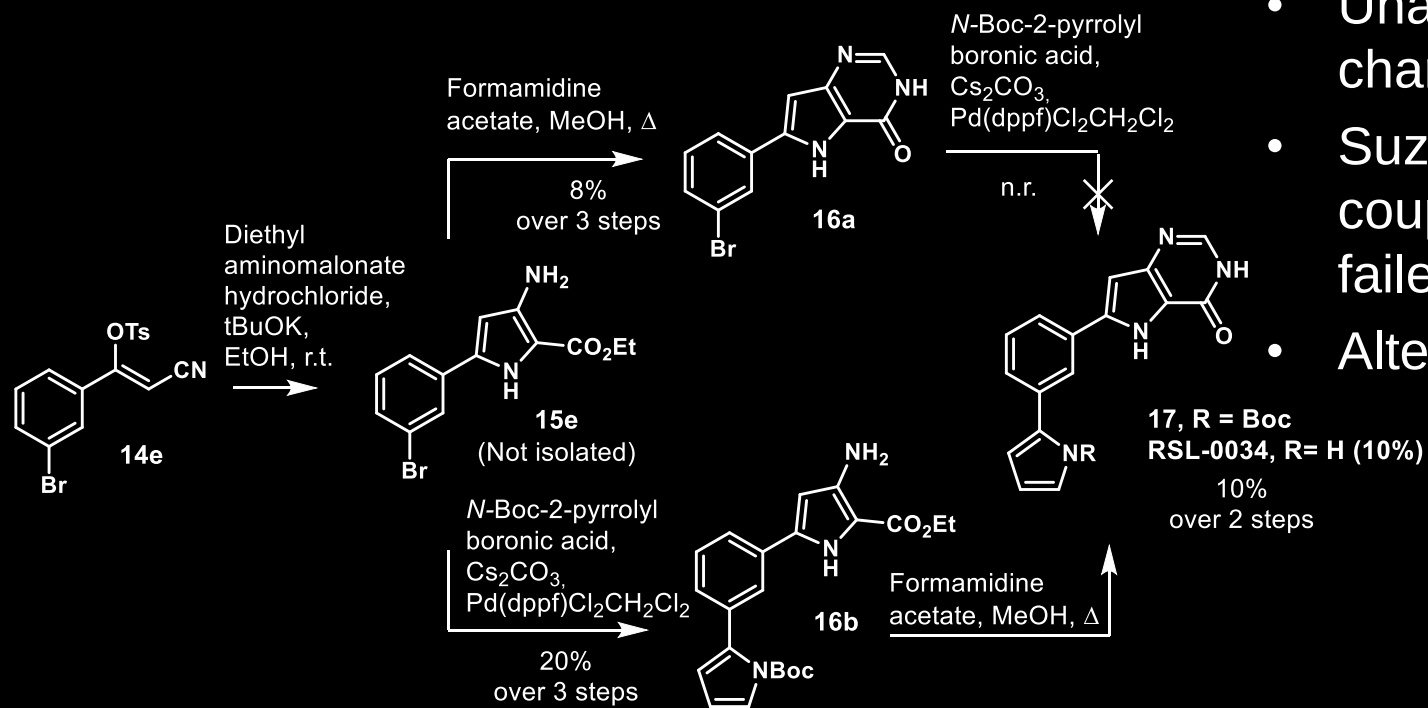
RSL-0031: ---Ph



- Similar synthesis
- Consecutive construction
- Various yields
- Purified



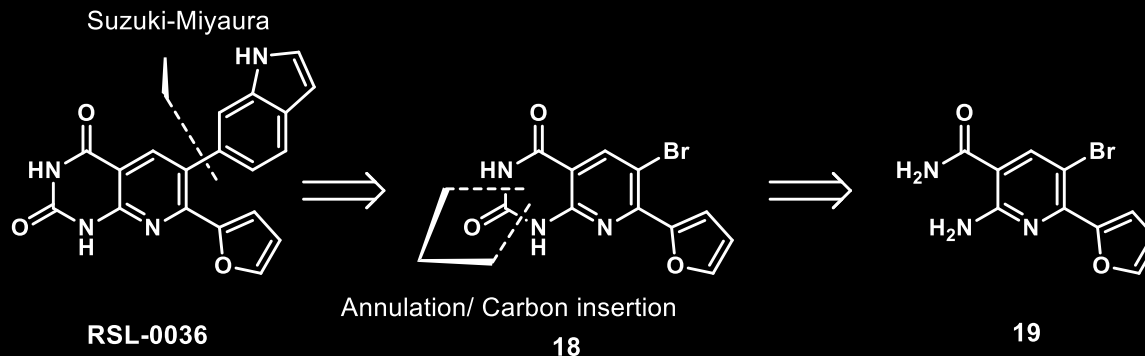
Synthesis of RSL-0034



- Unable to characterize **15e**
- Suzuki-Miyaura coupling of **16a** failed
- Alternative route



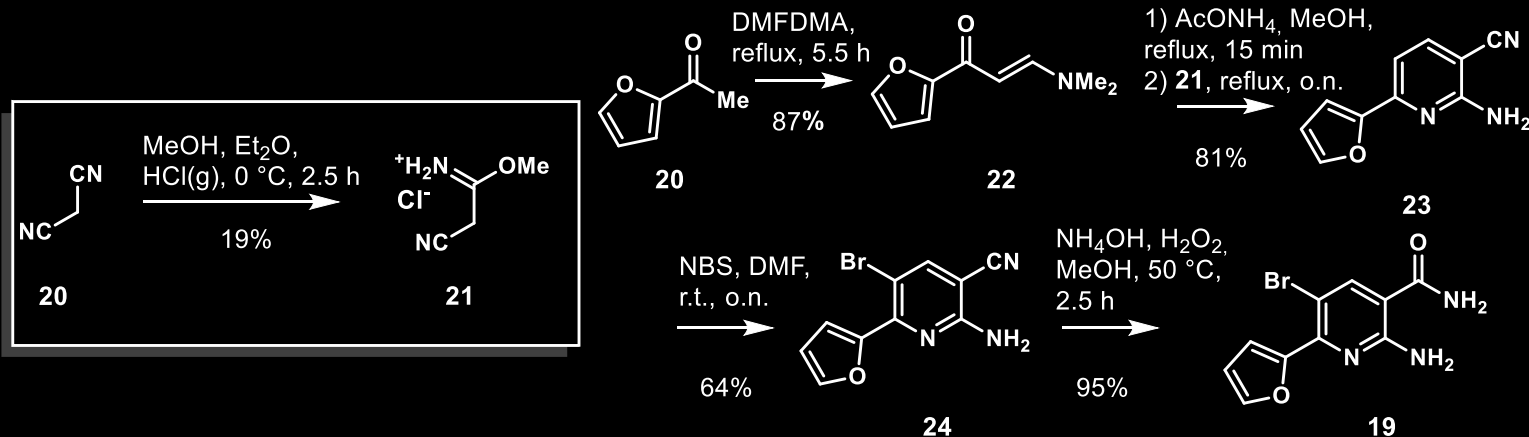
Synthesis of RSL-0036 I



- Analogue synthesis published, last step Suzuki-Miyaura



Synthesis of RSL-0036 II



Synthesis of RSL-0036 III

Method A:

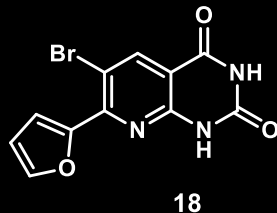
i) NaH, DMSO, 30 min, r.t.

ii) CDI, DMF, 2 h, 90 °C

Method B:

CDI, DMF, DBU, 110 °C

38%/49%



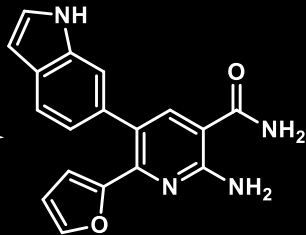
6-indolyl boronic acid
Cs₂CO₃, dioxane,
Pd(dppf)Cl₂•CH₂Cl₂

38%

19

6-indolyl boronic acid
Cs₂CO₃, dioxane,
Pd(dppf)Cl₂•CH₂Cl₂

85%



Method A:

i) NaH, DMSO, 30 min, r.t.

ii) CDI, DMF, 2 h, 90 °C

Method B:

CDI, DMF, DBU, 110 °C

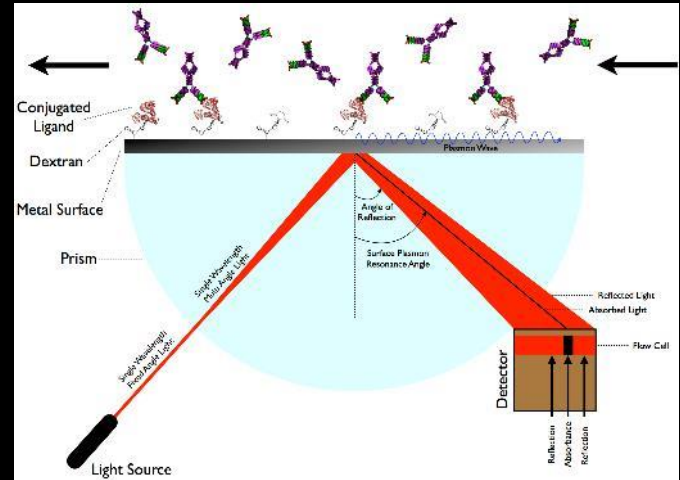


Structure-activity research



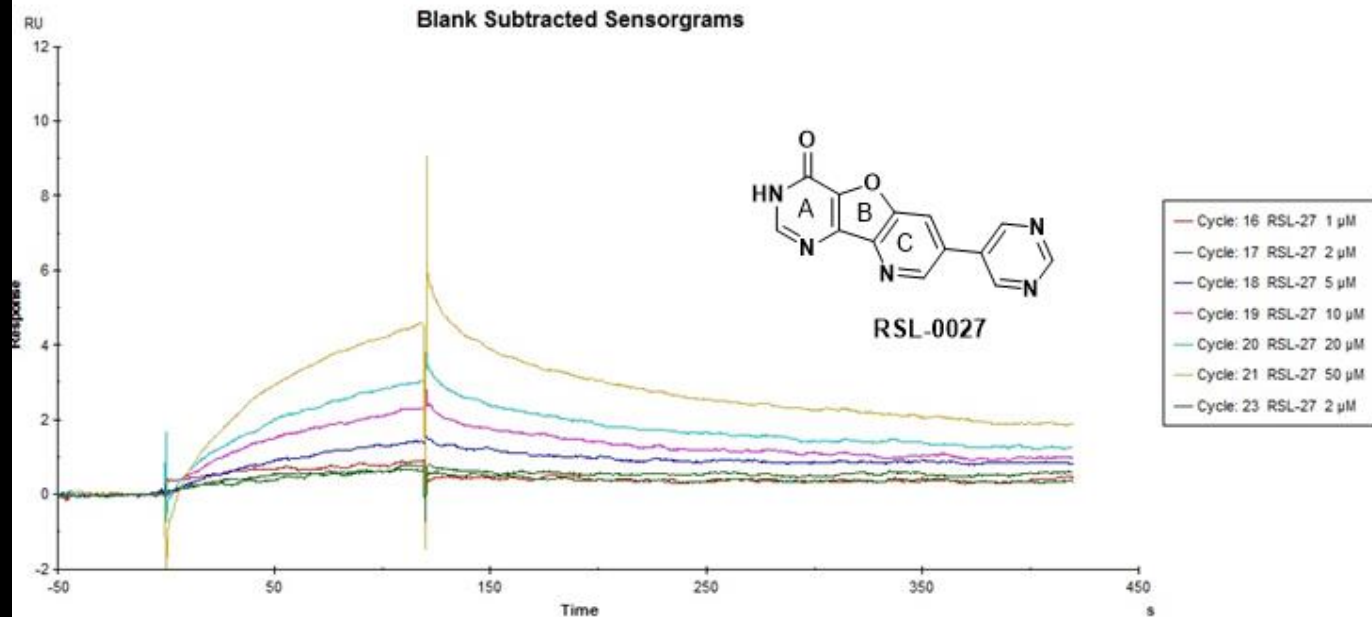
SPR assay & evaluation of activity

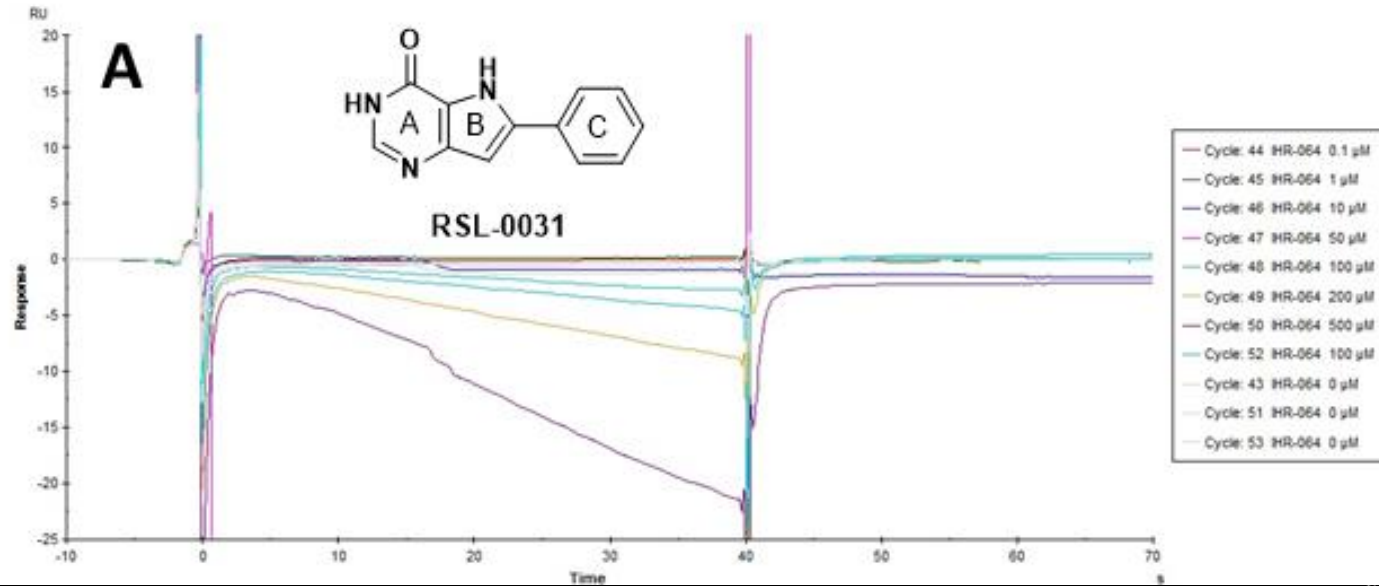
- Compounds were evaluated using SPR assay
- Experiments performed by Dr. Fahimeh Khorsand
- Immobilized FMN RS on a chip
- Analyte (compound) injected in increasing concentrations → Signal observed
- Affinity calculated

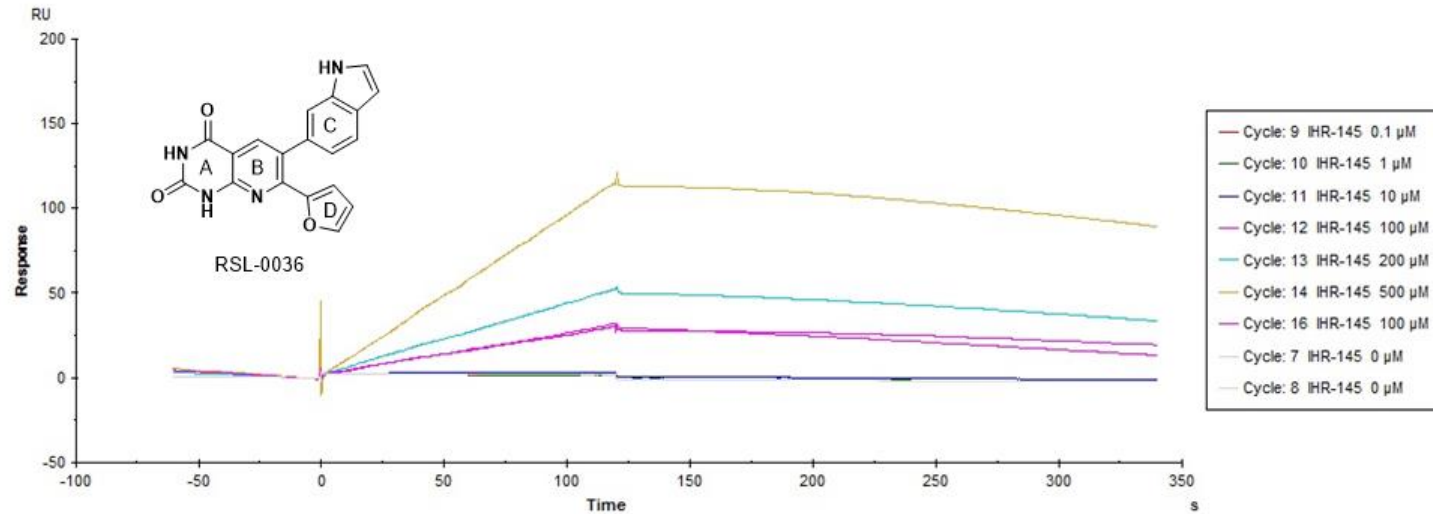


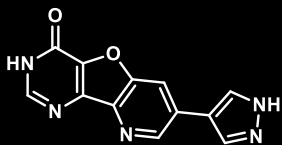
Sabban, Sari (2011) Development of an in vitro model system for studying the interaction of Equus caballus IgE with its high-affinity FcεRI receptor (PhD thesis), The University of Sheffield (CC-BY-SA-3.0)





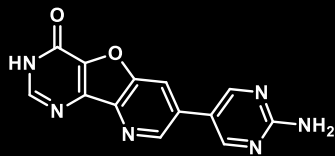






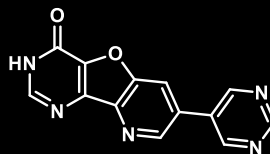
RSL-0029

Low solubility in DMSO



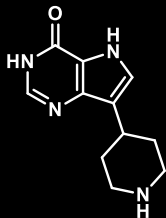
RSL-0026

$K_D = 2.2 \pm 0.7 \mu\text{M}$ ****
LE = 0.38 kcal·mol⁻¹ heavy atom⁻¹



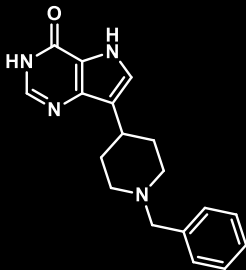
RSL-0027

$K_D = 5.9 \pm 0.3 \mu\text{M}$ ****
LE = 0.36 kcal·mol⁻¹ heavy atom⁻¹



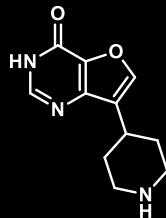
RSL-0012

$K_D = 187 \pm 60 \mu\text{M}$ ****
E = 0.34 kcal·mol⁻¹ heavy atom⁻¹



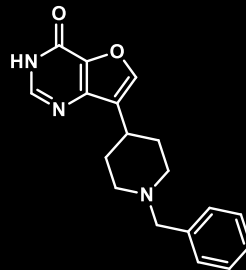
RSL-0007

Low solubility in DMSO



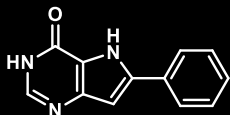
RSL-0015

$K_D = 133 \pm 18 \mu\text{M}$ ****
LE = 0.34 kcal·mol⁻¹ heavy atom⁻¹



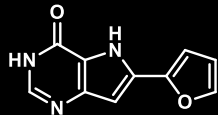
RSL-0014

Unable to fit



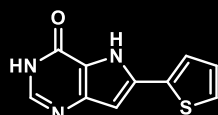
RSL-0031

No binding



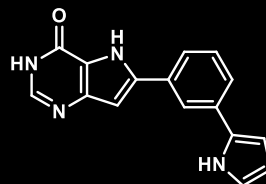
RSL-0032

$K_D = 120 \pm 30 \mu\text{M}$ ****
LE = 0.37 kcal·mol⁻¹ heavy atom⁻¹



RSL-0033

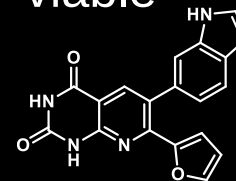
No binding



RSL-0034

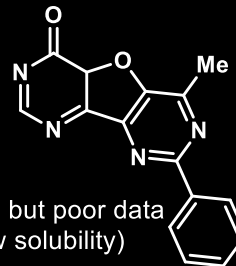
$K_D = 8.33 \pm 2.89 \mu\text{M}$ ****
LE = 0.34 kcal·mol⁻¹ heavy atom⁻¹

- Some compounds are troubled with solubility issues (Including RSL-0035)
- 3 compounds have $K_D < 10 \mu\text{M}$
- Scaffold seems viable



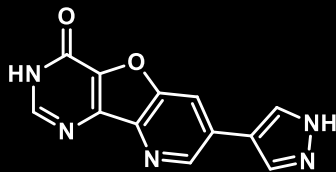
RSL-0036

Slow binding



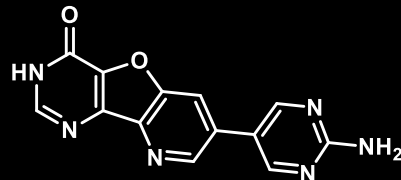
Activity, but poor data
(Low solubility)





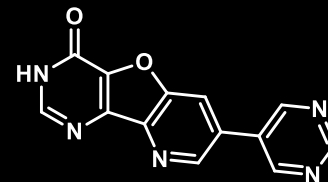
RSL-0029

Low solubility in DMSO



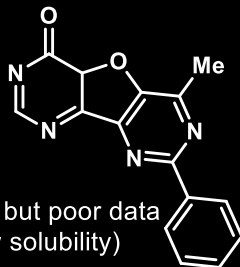
RSL-0026

$K_D = 2.2 \pm 0.7 \mu\text{M}$ ****
LE = 0.38 kcal·mol⁻¹·heavy atom⁻¹



RSL-0027

$K_D = 5.9 \pm 0.3 \mu\text{M}$ ****
LE = 0.36 kcal·mol⁻¹·heavy atom⁻¹



Activity, but poor data
(Low solubility)

- RSL-0035; poor candidate
- Similar scaffold still promising
- RSL-0026, RSL-0027: Highly promising activity
- Does the substituent contribute to affinity?

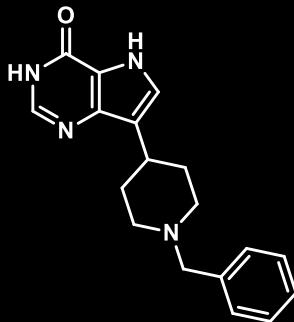




RSL-0012

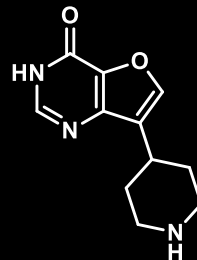
$K_D = 187+60 \mu\text{M}$ ****

$LE = 0.34 \text{ kcal}\cdot\text{mol}^{-1}\cdot\text{heavy atom}^{-1}$



RSL-007

Low solubility in DMSO



RSL-0015

$K_D = 133+18 \mu\text{M}$ ****

$LE = 0.34 \text{ kcal}\cdot\text{mol}^{-1}\cdot\text{heavy atom}^{-1}$

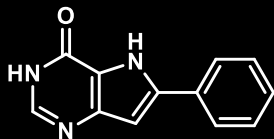


RSL-0014

Unable to fit

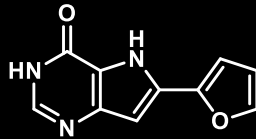
- RSL-0012, RSL-0015: promising start
- Resubmit RSL-0007, RSL-0014 as HCl salts?





RSL-0031

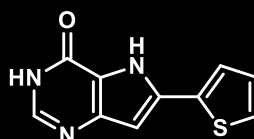
No binding



RSL-0032

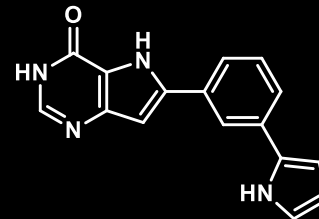
$K_D = 120 \pm 30 \text{ } \mu\text{M}$ ****

$LE = 0.37 \text{ kcal}\cdot\text{mol}\cdot\text{heavy atom}^{-1}$



RSL-0033

No binding



RSL-0034

$K_D = 8.33 \pm 2.89 \text{ } \mu\text{M}$ ****

$LE = 0.34 \text{ kcal}\cdot\text{mol}\cdot\text{heavy atom}^{-1}$

- RSL-0032, RSL-0034 show good activity
- Despite different K_D , LE similar



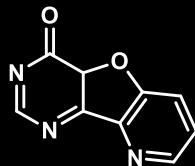
Conclusion

- Created a druggability predictor compliant with RNA
- Proposed a series of ligands for the FMN riboswitch
- Synthesized and purified said ligands + previous hit
- Validated binding using SPR
- Found previous hit to be a poor candidate



Further work?

- Crystal structure to validate binding mode
- Validate central scaffold



- Overcome slow kinetics of RSL-0036



Acknowledgements

- Grants: UiB, Norsk Forskningsråd
- Professor Ruth Brenk, Professor Bengt Erik Haug
- Dr. Fahimeh Khorsand for SPR experiments and analyses
- Dr. Muhammed Zeeshan, Valeriia Burova, Heidi Kristine Vintermyhr, Andrea Osvoll Årdal
- Dr. Bjarte Holmelid for HRMS analyses
- Professor Nils Åge Frøystein, Professor Jarl Underhaug, Jose Carlos Reyes Guerrero for assistance with NMR
- Everyone in the Brenk and the Haug labs



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@illimar



**Thank you all for
listening!**





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